Convertible Car Mechanism

ENME 473: Fundamentals of Kinematics and Dynamics of Machines

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1 Identification of Links and Joints

For the convertible car top mechanism, all links and joints were identified and labelled. The results can be seen below in Figure 1.

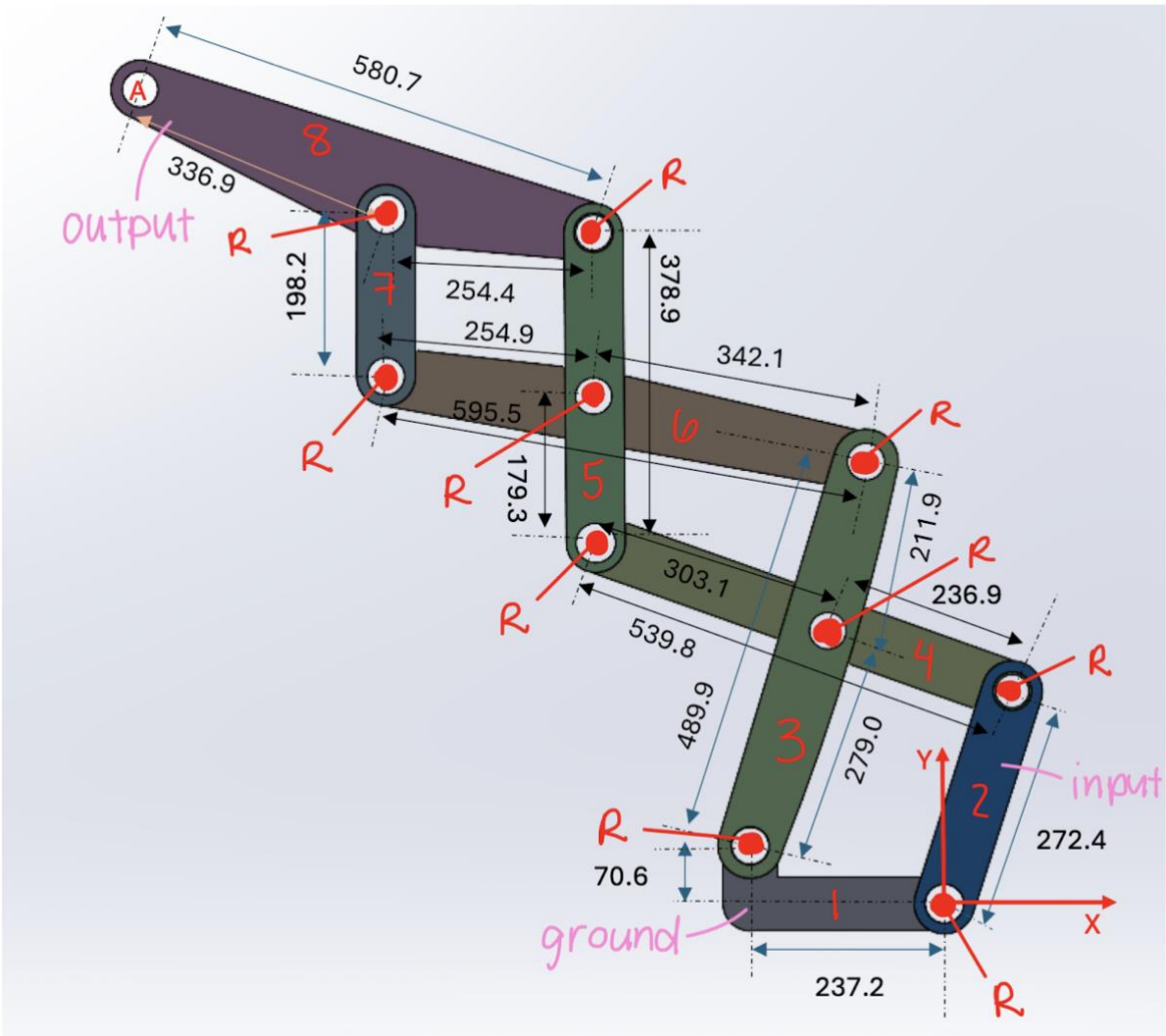


Figure 1: Annotated Diagram of all Links and Joints

Using the information from Figure 1 above, the mobility of the mechanism can be determined using Equation 1 below.

$$m = 3(n-1) - 2(J_1) - J_2$$

$$m = 3(8-1) - 2(10) - (0) \ = \ 1$$

With a total of 8 links, 10 J1 joints and 0 J2 joints, the mobility of the mechanism was found to be 1.

2 Vector Loop Analysis

This mechanism can then be broken down into three separate vector loops which will help to simplify the analysis.

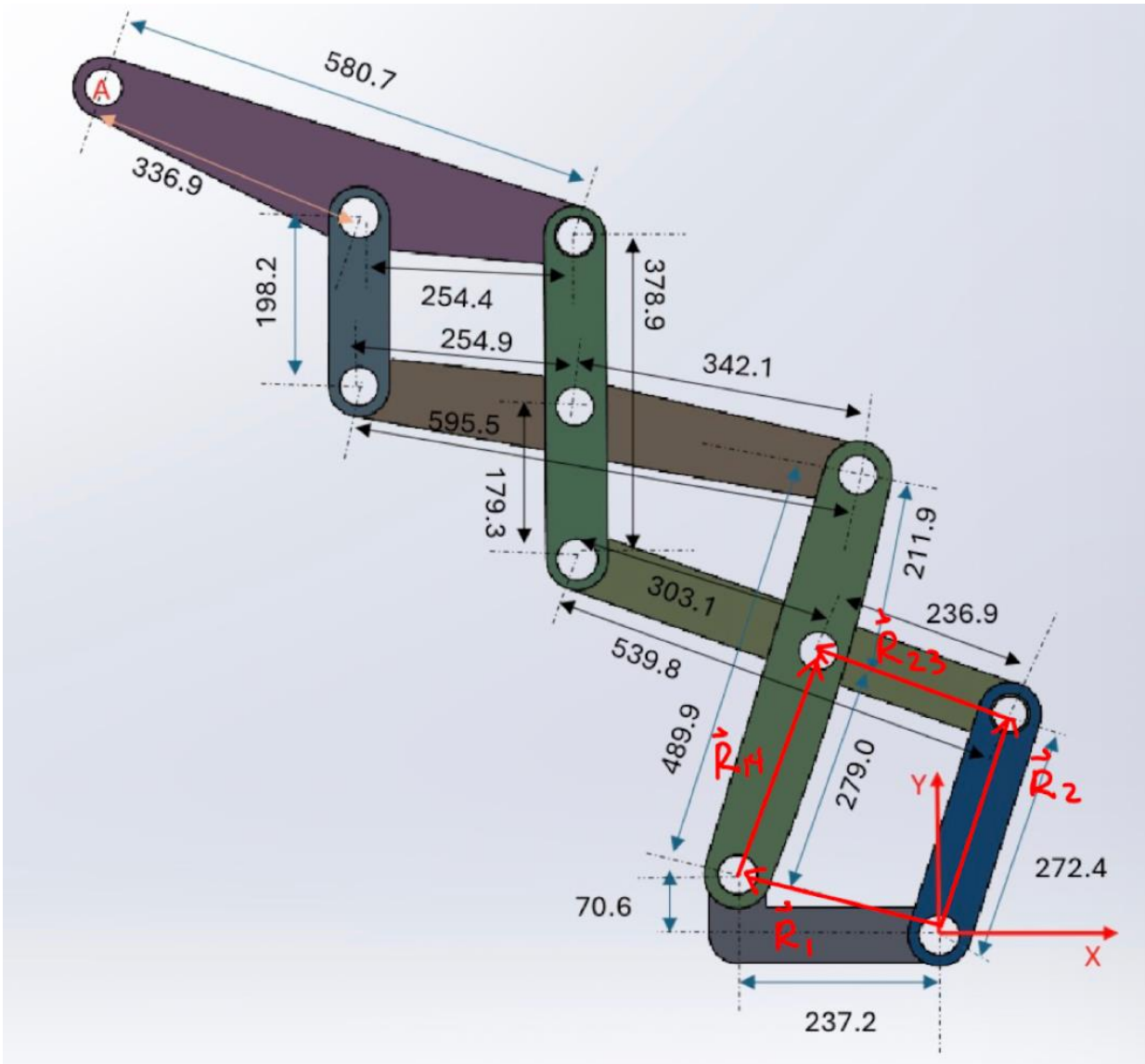


Figure 2: Annotated Diagram of Loop 1

Vectors: $\overline{R}_1$, $\overline{R}_{14}$, $\overline{R}_{23}$, $\overline{R}_2$

Knowns: R_1, R_{14}, R_{23}, R_2, θ_1, θ_2

Unknowns: θ_{14}, θ_{23}

Vector Equation:

$$\overline{R}_2 + \overline{R}_{23} - \overline{R}_{14} - \overline{R}_1 = 0$$

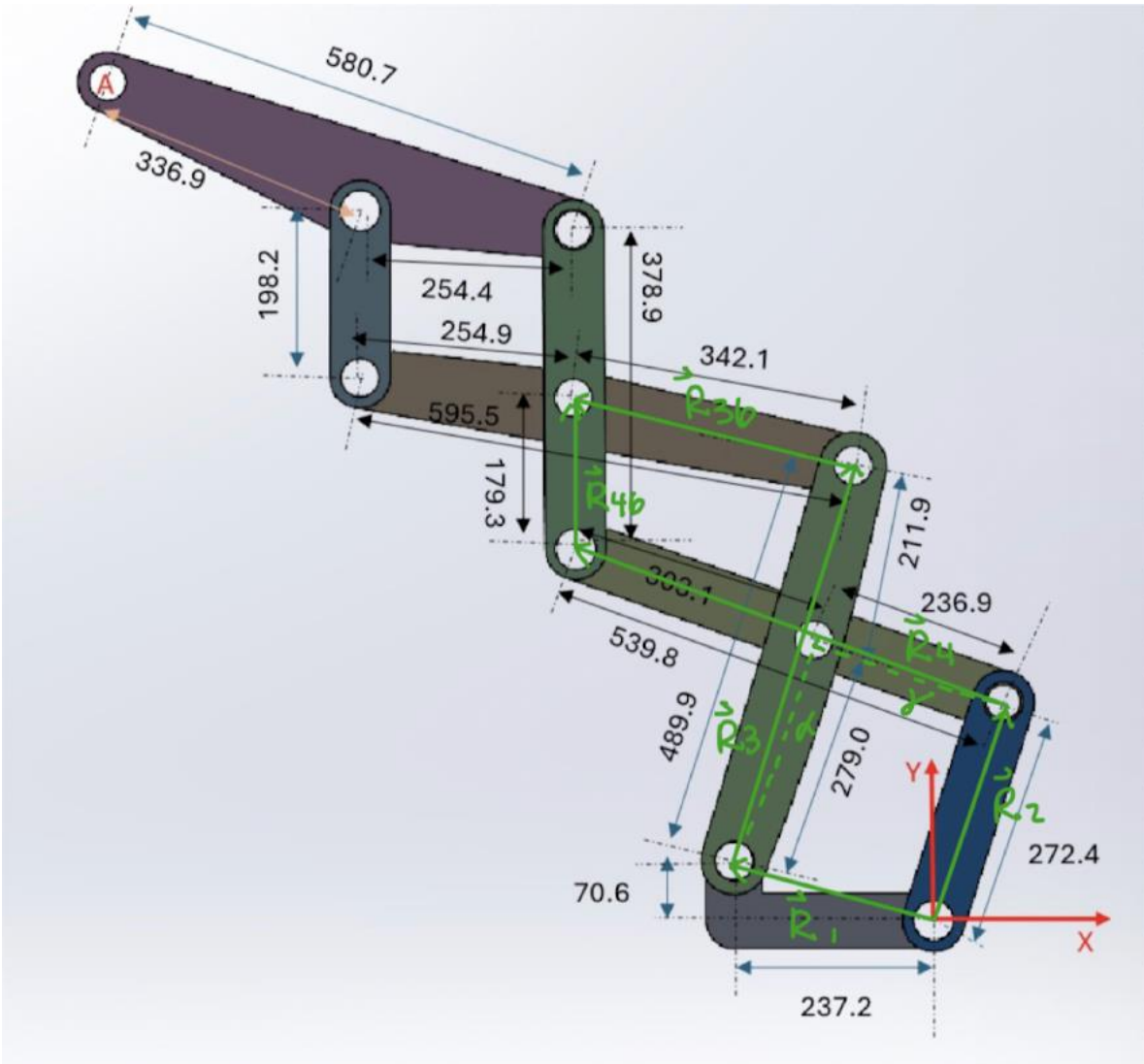


Figure 3: Annotated Diagram of Loop 2

Vector: $\overline{R}_1$, $\overline{R}_2$, $\overline{R}_3$, $\overline{R}_4$, $\overline{R}_{36}$, $\overline{R}_{46}$

Knowns: R_1, R_2, R_3, R_4, R_{36}, R_{46}, θ_1, θ_2

Unknowns: θ_{36}, θ_{46}

Constraints:

C_1: $\theta_3 = \theta_{14} + \alpha$

C_2: $\theta_4 = \theta_{23} + \gamma$

Vector Equation:

$$\overline{R}_2 + \overline{R}_4 + \overline{R}_{46} - \overline{R}_{36} - \overline{R}_3 - \overline{R}_1 = 0$$

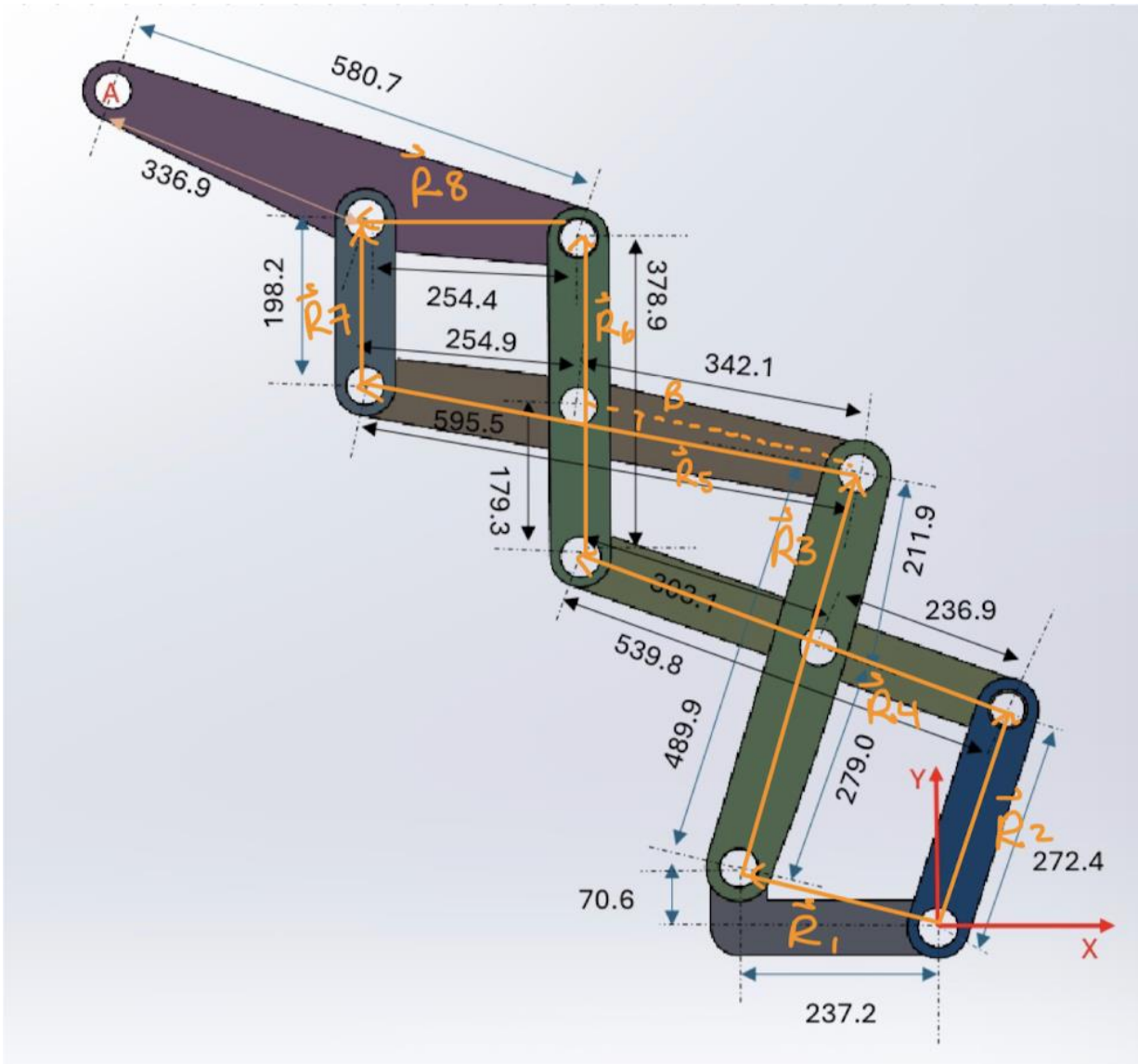


Figure 4: Annotated Diagram of Loop 3

Vectors: $\overline{R}_1$, $\overline{R}_2$, $\overline{R}_3$, $\overline{R}_4$, $\overline{R}_5$, $\overline{R}_6$, $\overline{R}_7$, $\overline{R}_8$

Knowns: R_1, R_2, R_3, R_4, R_5, R_6, R_7, R_8, θ_1, θ_2

Unknowns: θ_7, θ_8

Constraints:

C_3: $\theta_6 = \theta_{46}$

C_4: $\theta_5 = \theta_{36} + \beta$

Vector Equation:

$$\overline{R}_2 + \overline{R}_4 + \overline{R}_6 + \overline{R}_8 - \overline{R}_7 - \overline{R}_5 - \overline{R}_3 - \overline{R}_1 = 0$$

For the following analysis of the mechanism, the lengths of joints were left as variables to reduce noise. Tabulated below are the geometric results of the length values.

Table 1: Length Values Used for Final Calculations

	R_1	R_2	R_{23}	R_{14}	R_3	R_4	R_{36}	R_{46}	R_5	R_6	R_7	R_8	R_A
Length (mm)	247.5	272.4	236.9	279.0	489.9	539.8	342.1	179.3	595.5	378.9	198.2	254.4	336.9

The constraint angles will also be left as variables. The angle values were solved using the cosine law and can be seen below in Table 2.

Table 2: Constraint Angle Values Used for Final Calculations

	α	γ	β
Angle (degrees)	3.2	1.8	3.5

3 Posture Analysis

3.1 Discretization of Variables

To summarize the results of the vector loops above, the three vector loop equations can be seen below in Table 1.

Table 3: Summarized Vector Loop Equations

Vector Equations:
$\overline{R}_2 + \overline{R}_{23} - \overline{R}_{14} - \overline{R}_1 = 0$
$\overline{R}_2 + \overline{R}_4 + \overline{R}_{46} - \overline{R}_{36} - \overline{R}_3 - \overline{R}_1 = 0$
$\overline{R}_2 + \overline{R}_4 + \overline{R}_6 + \overline{R}_8 - \overline{R}_7 - \overline{R}_5 - \overline{R}_3 - \overline{R}_1 = 0$

Each of these equations can then be split into the force equations for both the x and y directions.

For the first loop:

$$f_x = R_2 cos\ \theta_2 + R_{23} cos\ \theta_{23} - R_{14} cos\ \theta_{14} - R_1 cos\ \theta_1$$

$$f_y = R_2 sin\ \theta_2 + R_{23} sin\ \theta_{23} - R_{14} sin\ \theta_{14} - R_1 sin\ \theta_1$$

For the second loop:

$$f_x = R_2 cos\ \theta_2 + R_4 cos\ \theta_4 + R_{46} cos\ \theta_{46} - R_{36} cos\ \theta_{36} - R_3 cos\ \theta_3 - R_1 cos\ \theta_1$$

$$f_y = R_2 sin\ \theta_2 + R_4 sin\ \theta_4 + R_{46} sin\ \theta_{46} - R_{36} sin\ \theta_{36} - R_3 sin\ \theta_3 - R_1 sin\ \theta_1$$

For the third loop:

$$f_y = R_2 cos\ \theta_2 + R_4 cos\ \theta_4 + R_6 cos\ \theta_6 + R_8 cos\ \theta_8 - R_7 cos\ \theta_7 - R_5 cos\ \theta_5 - R_3 cos\ \theta_3 - R_1 cos\ \theta_1$$

$$f_y = R_2 sin\,\theta_2 + R_4 sin\,\theta_4 + R_6 sin\,\theta_6 + R_8 sin\,\theta_8 - R_7 sin\,\theta_7 - R_5 sin\,\theta_5 \; - R_3 sin\,\theta_3 - R_1 sin\,\theta_1$$

To reduce the number of unknowns, the constrain equations can be used to limit the 6 equations to 6 unknowns.

$$C_1\colon\ \theta_3 = \theta_{14} + \alpha$$
$$C_2\colon\ \theta_4 = \theta_{23} \; + \; \gamma$$
$$C_3\colon\ \theta_6 = \theta_{46}$$
$$C_4\colon\ \theta_5 = \theta_{36} + \beta$$

The length and angle of the ground link can also be calculated.

$$R_1 = \sqrt{(237.2)^2 + (70.6)^2} = 247.5\ mm$$

$$180 - \theta_1 = \tan^{-1}\left(\frac{70.6}{237.2}\right) = 16.6°$$

$$\theta_1 = 163.4°$$

The Jacobian of the 6 equations above is then given by:

$-R_{23} sin\,\theta_{23}$	$R_{14} sin\,\theta_{14}$	0	0	0	0
$R_{23} cos\,\theta_{23}$	$-R_{14} cos\,\theta_{14}$	0	0	0	0
$-R_4 sin\,(\theta_{23} + \gamma)$	$R_3 sin\,(\theta_{14} + \alpha)$	$-R_{46} sin\,\theta_{46}$	$R_{36} sin\,\theta_{36}$	0	0
$R_4 cos(\theta_{23} + \gamma)$	$-R_3 cos\,(\theta_{14} + \alpha)$	$R_{46} cos\,\theta_{46}$	$-R_{36} cos\,\theta_{36}$	0	0
$-R_4 sin\,(\theta_{23} + \gamma)$	$R_3 sin\,(\theta_{14} + \alpha)$	$-R_6 sin\,\theta_{46}$	$R_5 sin\,(\theta_{36} + \beta)$	$R_8 sin\,\theta_8$	$-R_7 sin\,\theta_7$
$R_4 cos(\theta_{23} + \gamma)$	$-R_3 cos\,(\theta_{14} + \alpha)$	$R_6 cos\,\theta_{46}$	$-R_5 cos\,(\theta_{36} + \beta)$	$-R_8 cos\,\theta_8$	$R_7 cos\,\theta_7$

Solving in MATLAB gives the following graph of link angles compared to the input angle for a range of 0°120°.

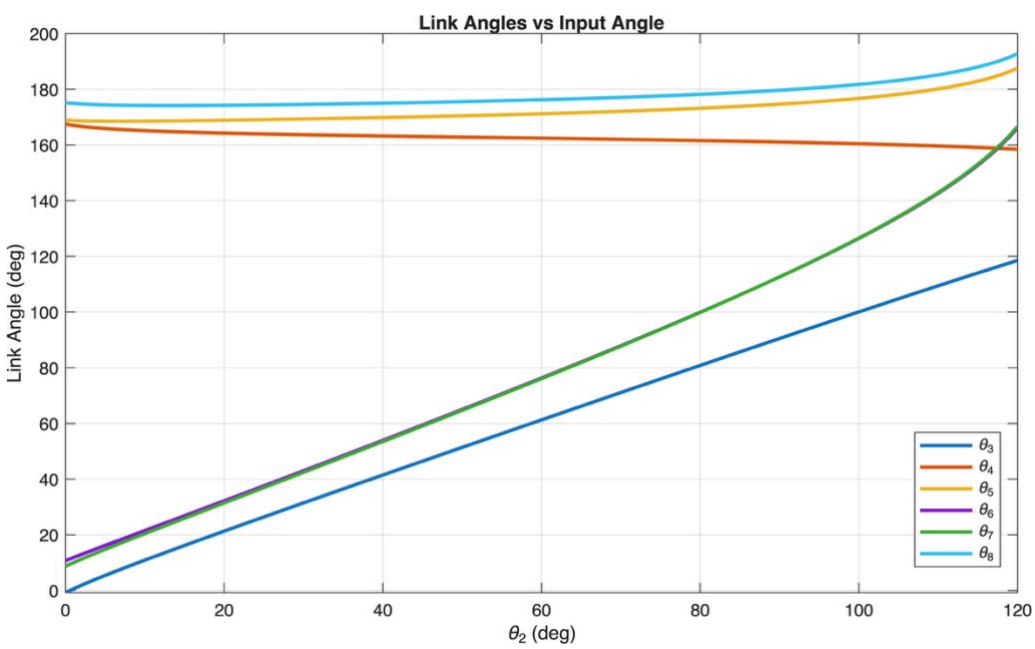


Figure 5: Graphical Results of Numerical Solution

The tabular results for the numerical solution at selected input angles can be seen below in Table 3. For full tabular results, refer to Appendix A.

Table 4: Tabular Results of Numerical Solution

$\boldsymbol{\theta_2}$ [degrees]	$\boldsymbol{\theta_3}$ [degrees]	$\boldsymbol{\theta_4}$ [degrees]	$\boldsymbol{\theta_5}$ [degrees]	$\boldsymbol{\theta_6}$ [degrees]	$\boldsymbol{\theta_7}$ [degrees]	$\boldsymbol{\theta_8}$ [degrees]
0	-0.75	167.48	168.90	10.73	8.72	175.19
30	31.46	163.69	169.33	42.99	42.44	174.55
60	61.27	162.44	171.21	76.31	76.10	176.25
90	90.50	161.06	174.65	112.62	112.64	179.64
120	118.54	158.49	187.56	166.14	166.64	192.75

3.2 Four Bar Linkage Analysis

In the given mechanism, it can be noted that links 1, 2, 3 and 4 make up a four-bar linkage. This is the same linkage that is shown by vector loop 1. To confirm the results obtained from MATLAB, the four-bar linkage was solved analytically using closed-form solution methods at two different input angles.

To begin the analytical solution, start with the vector loop equation for the four-bar linkage.

$$\overline{R}_2 + \overline{R}_{23} - \overline{R}_{14} - \overline{R}_1 = 0 \tag{1}$$

This equation can then be split into force equations for both x and y directions.

$$f_x = R_2 cos\ \theta_2 + R_{23} cos\ \theta_{23} - R_{14} cos\ \theta_{14} - R_1 cos\ \theta_1 = 0$$

$$f_y = R_2 sin\ \theta_2 + R_{23} sin\ \theta_{23} - R_{14} sin\ \theta_{14} - R_1 sin\ \theta_1 = 0$$

These equations can then be rearranged as follows.

$$R_{14} cos\theta_{14} = R_2 cos\theta_2 + R_{23} cos\theta_{23} - R_1 cos\theta_1 \tag{2}$$

$$R_{14} sin\theta_{14} = R_2 sin\theta_2 + R_{23} sin\theta_{23} - R_1 sin\theta_1 \tag{3}$$

Begin solving by calculating $(2)^2 + (3)^2$.

$$\begin{aligned} R_{14}^2(cos^2\theta_{14} + sin^2\theta_{14}) \\ &= R_2^2 cos^2\theta_2 + R_{23}^2 cos^2\theta_{23} + R_1^2 cos^2\theta_1 + 2R_2R_{23} cos\theta_2 cos\theta_{23} - 2R_2R_1 cos\theta_2 cos\theta_1 \\ &- 2R_{23}R_1 cos\theta_{23} cos\theta_1 + R_2^2 sin^2\theta_2 + R_{23}^2 sin^2\theta_{23} + R_1^2 sin^2\theta_1 + 2R_2R_{23} sin\theta_2 sin\theta_{23} \\ &- 2R_2R_1 sin\theta_2 sin\theta_1 - 2R_{23}R_1 sin\theta_{23} sin\theta_1 \end{aligned}$$

This can then be rearranged to be in the format $C = Acos\theta_{23} + Bsin\theta_{23}$.

$$\begin{aligned} R_{14}^2 - R_{14}^2 - R_{14}^2 - R_{14}^2 + 2R_2R_1 cos\theta_2 cos\theta_1 + 2R_2R_1 sin\theta_2 sin\theta_1 \\ = 2R_{23}(R_2 cos\theta_2 - R_1 cos\theta_1) cos\theta_{23} + 2R_{23}(R_2 sin\theta_2 - R_1 sin\theta_1) sin\theta_{23} \end{aligned}$$

This gives coefficients A, B and C as:

$$A = 2R_{23}(R_2 cos\theta_2 - R_1 cos\theta_1)$$

$$B = 2R_{23}(R_2 sin\theta_2 - R_1 sin\theta_1)$$

$$C = R_{14}^2 - R_{14}^2 - R_{14}^2 - R_{14}^2 + 2R_2R_1 cos\theta_2 cos\theta_1 + 2R_2R_1 sin\theta_2 sin\theta_1$$

Using these constants, the values of θ_{23} and θ_{14} can be determined using the following equations:

$$\theta_{23} = \emptyset \pm cos^{-1}\left(\frac{C}{\sqrt{A^2 + B^2}}\right)$$

$$\emptyset = tan^{-1}\left(\frac{B}{A}\right)$$

$$\theta_{14} = cos^{-1}\left(\frac{R_2 cos\theta_2 + R_{23} cos\theta_{23} - R_1 cos\theta_1}{R_{14}}\right)$$

Table 2 below shows the results of the analytical solution for input angles of 30° and 90°

Table 5: Results from Analytical Solution of the Four-Bar Linkage

$\boldsymbol{\theta_2}$	A	B	C	$\boldsymbol{\theta_{14}}$	$\boldsymbol{\theta_{23}}$
30°	224157	31081	-206413	31.46°	163.69°
90°	112385	95613	-75268	90.50°	161.06°

The results of the analytical solution match the values from the MATLAB code, therefore verifying the results obtained using the Newton-Raphson method.

3.3 Point Path Traced by Pin A

To solve for each degree increment the Newton-Raphson method was used to solve six kinematic loop closure equations which represent the three total loops in the system. A Jacobian matrix was then made containing the partial derivatives of the equations with respect to the unknown angles in order to continuously update the angle estimates until the values were within the desired error range and converging.

Once a correct set of link angles is found, the position of Pin A was calculated by summing the x and y components of each vector in the kinematic chain from the origin to Pin A. The position of Pin A was stored for every input in order to plot the position of Pin A during the motion of the mechanism as seen in Figure 6.

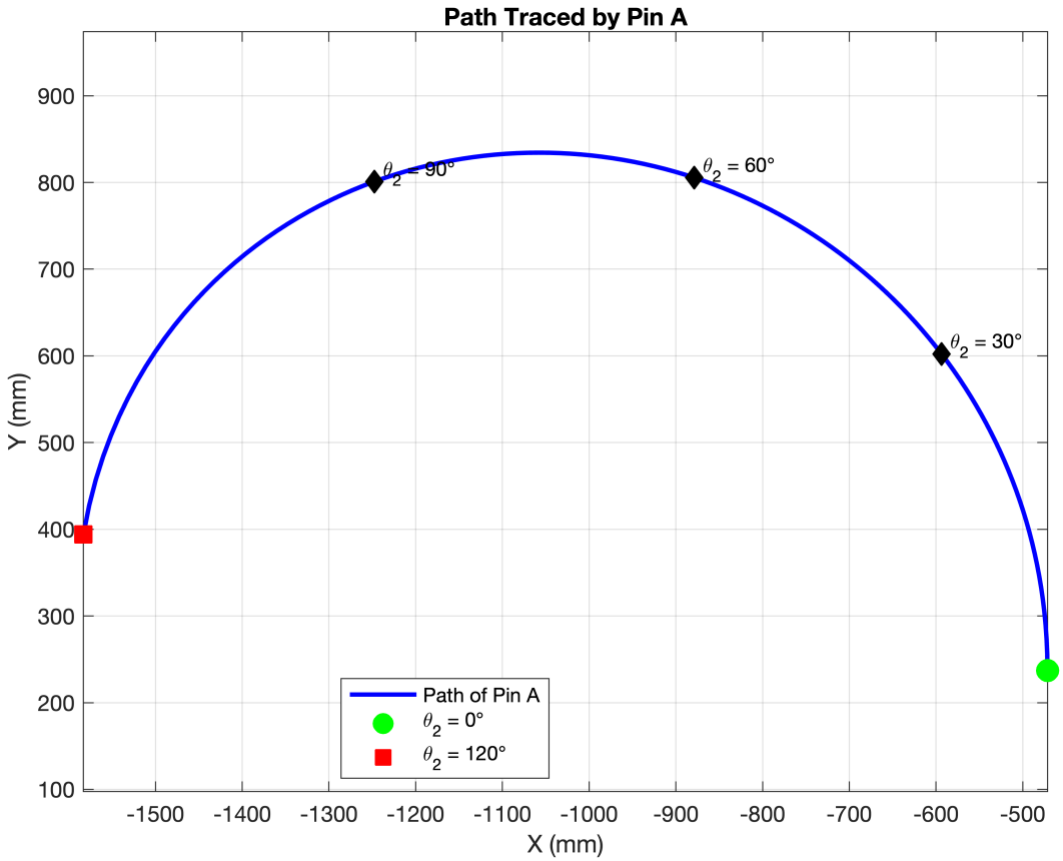


Figure 6: Path Traced by Pin A

Below in Table 5, are the main input angles with the fully tabulated results in 6.2

Table 6: Tabulated Results of Path Traced by Pin A

$\boldsymbol{\theta_2}$ [degrees]	**X – Position of A [mm]**	**Y – Position of A [mm]**
0	-471.50	237.11
30	-593.66	602.25
60	-878.80	805.59
90	-1247.58	801.03
120	-1583.01	394.08

4 Velocity and Acceleration Analysis

To determine the angular velocities and accelerations of each link, the position equations from Section 3 are differentiated with respect to time. The same Jacobian matrix obtained during the posture analysis is reused for both the velocity and acceleration solutions. An input angular velocity of 1 rad/s with zero angular acceleration is assumed for link 2.

$$\omega_2 = 1\, rad/s, \;\; \alpha_2 = \; 0\, rad/s^2$$

4.1 Velocity Analysis

The velocity equations are obtained by differentiating the six position equations with respect to time. Differentiating gives:

$$\frac{d}{dt}[f(\theta_2, x)] = \frac{\partial f}{\partial \theta_2}\omega_2 + \frac{\partial f}{\partial x}\dot{x} = 0$$

This can then be rearranged to solve for the unknown angular velocities:

$$J\dot{x} = -\frac{\partial f}{\partial \theta_2}\omega_2 = b\omega_2$$

J represents the same Jacobian as the posture analysis, and the unknow velocity vector is:

$$\dot{x} = \left[\dot{\theta}_{23}, \dot{\theta}_{14}, \dot{\theta}_{46}, \dot{\theta}_{36}, \dot{\theta}_8, \dot{\theta}_7\right]^T$$

First loop:

$$-R_{23}\sin(\theta_{23})\,\dot{\theta}_{23} + R_{14}\sin(\theta_{14})\,\dot{\theta}_{14} = R_2\sin(\theta_2)\,\omega_2$$

$$R_{23}\cos(\theta_{23})\,\dot{\theta}_{23} - R_{14}\cos(\theta_{14})\,\dot{\theta}_{14} = -R_2\cos(\theta_2)\,\omega_2$$

Second loop:

$$-R_4\sin(\theta_{23})\,\dot{\theta}_{23} + R_3\sin(\theta_{14}+\alpha)\,\dot{\theta}_{14} - R_{46}\sin(\theta_{46})\,\dot{\theta}_{46} + R_{36}\sin(\theta_{36})\,\dot{\theta}_{36} = R_2\sin(\theta_2)\,\omega_2$$

$$-R_4\cos(\theta_{23})\,\dot{\theta}_{23} - R_3\cos(\theta_{14}+\alpha)\,\dot{\theta}_{14} + R_{46}\cos(\theta_{46})\,\dot{\theta}_{46} - R_{36}\cos(\theta_{36})\,\dot{\theta}_{36} = -R_2\cos(\theta_2)\,\omega_2$$

Third loop:

$$-R_4\sin(\theta_{23})\,\dot{\theta}_{23} + R_3\sin(\theta_{14}+\alpha)\,\dot{\theta}_{14} - R_6\sin(\theta_{46})\,\dot{\theta}_{46} + R_5\sin(\theta_{36}+\beta)\,\dot{\theta}_{36}$$

$$-R_8\sin(\theta_8)\,\dot{\theta}_8 + R_7\sin(\theta_7)\,\dot{\theta}_7 = R_2\sin(\theta_2)\,\omega_2$$

$$R_4\cos(\theta_{23})\,\dot{\theta}_{23} - R_3\cos(\theta_{14}+\alpha)\,\dot{\theta}_{14} - R_6\cos(\theta_{46})\,\dot{\theta}_{46} - R_5\cos(\theta_{36}+\beta)\,\dot{\theta}_{36}$$

$$+R_8\cos(\theta_8)\,\dot{\theta}_8 - R_7\cos(\theta_7)\,\dot{\theta}_7 = -R_2\cos(\theta_2)\,\omega_2$$

Vector b:

$$b = [R_2\sin(\theta_2), -R_2\cos(\theta_2), R_2\sin(\theta_2), -R_2\cos(\theta_2), R_2\sin(\theta_2), -R_2\cos(\theta_2),]^T$$

Using the constrain equations, we can determine the angular velocities of the links, that is to say:

$$\omega_3 = \dot{\theta}_{14}$$
$$\omega_4 = \dot{\theta}_{23}$$
$$\omega_5 = \dot{\theta}_{36}$$
$$\omega_6 = \dot{\theta}_{46}$$

$$\alpha_3 = \ddot{\theta}_{14}$$
$$\alpha_3 = \ddot{\theta}_{23}$$
$$\alpha_3 = \ddot{\theta}_{36}$$
$$\alpha_3 = \ddot{\theta}_{46}$$

For a range of 0º to 120º:

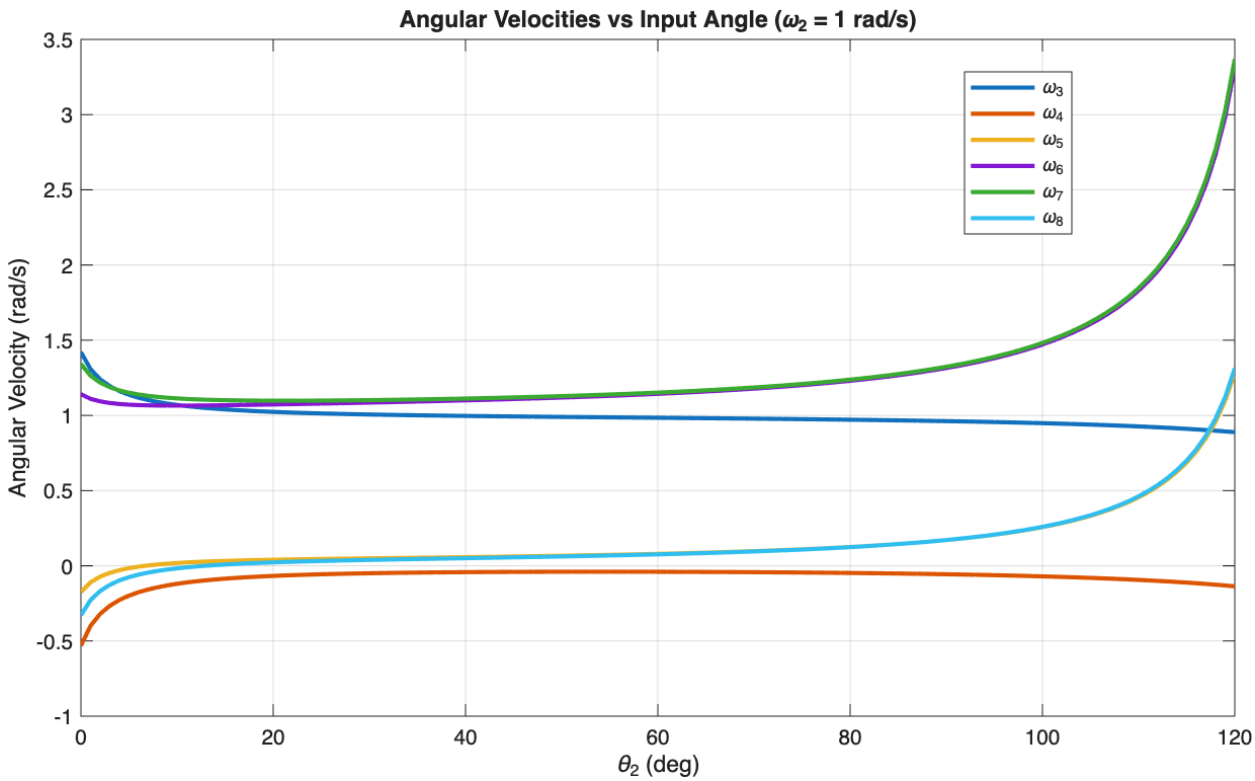


Figure 7: Angular Velocities vs. Input Angle

The tabular results are presented in the table below. Complete table in appendix.

Table 7: Angular Velocities at Select Input Angles (rad/s)

θ_2 [deg]	ω_3	ω_4	ω_5	ω_6	ω_7	ω_8
0	1.42	-0.53	-0.17	1.14	1.34	-0.33
30	1.01	-0.05	0.05	1.09	1.10	0.04
60	0.98	-0.04	0.08	1.14	1.15	0.08
90	0.96	-0.06	0.17	1.32	1.32	0.17
120	0.89	-0.14	1.28	3.32	3.37	1.31

4.2 Angular Acceleration Analysis

Acceleration equations are obtained by differentiating the velocity equations once more with respect to time:

$$J\ddot{x} + \dot{J}\dot{x} + \frac{\partial^2 f}{\partial \theta_2^2}\omega_2^2 + \frac{\partial f}{\partial x}\alpha_2 = 0$$

Knowing that $\alpha_2 = 0$,

$$J\ddot{x} = c$$

Where the unknown acceleration vector is:

$$\ddot{x} = \left[\ddot{\theta}_{23}, \ddot{\theta}_{14}, \ddot{\theta}_{46}, \ddot{\theta}_{36}, \ddot{\theta}_8, \ddot{\theta}_7\ \right]^T$$

The vector c represents the centripetal acceleration terms that come from differentiating the Jacobian, and these are computed from the known angular positions and velocities.

First loop:

$$c_1 = R_2 \cos(\theta_2)\, \omega_2^2 + R_{23} \cos(\theta_{23})\, \dot{\theta}_{23}^2 - R_{14} \cos(\theta_{14})\, \dot{\theta}_{14}^2$$

$$c_2 = R_2 \sin(\theta_2)\, \omega_2^2 + R_{23} \sin(\theta_{23})\, \dot{\theta}_{23}^2 - R_{14} \sin(\theta_{14})\, \dot{\theta}_{14}^2$$

Second loop:

$$c_3 = R_2 \cos(\theta_2)\, \omega_2^2 + R_4 \cos(\theta_{23})\, \dot{\theta}_{23}^2 + R_{46} \cos(\theta_{46})\, \dot{\theta}_{46}^2 - R_{36} \cos(\theta_{36})\, \dot{\theta}_{36}^2 - R_3 \cos(\theta_{14} + \alpha)\, \dot{\theta}_{14}^2$$

$$c_4 = R_2 \sin(\theta_2)\, \omega_2^2 + R_4 \sin(\theta_{23})\, \dot{\theta}_{23}^2 + R_{46} \sin(\theta_{46})\, \dot{\theta}_{46}^2 - R_{36} \sin(\theta_{36})\, \dot{\theta}_{36}^2 - R_3 \sin(\theta_{14} + \alpha)\, \dot{\theta}_{14}^2$$

Third loop:

$$c_5 = R_2 \cos(\theta_2)\, \omega_2^2 + R_4 \cos(\theta_{23})\, \dot{\theta}_{23}^2 + R_6 \cos(\theta_{46})\, \dot{\theta}_{46}^2 + R_8 \cos(\theta_8)\, \dot{\theta}_8^2$$

$$-R_7 \cos(\theta_7)\, \dot{\theta}_7^2 - R_5 \cos(\theta_{36} + \beta)\, \dot{\theta}_{36}^2 - R_3 \cos(\theta_{14} + \alpha)\, \dot{\theta}_{14}^2$$

$$c_6 = R_2 \sin(\theta_2)\, \omega_2^2 + R_4 \sin(\theta_{23})\, \dot{\theta}_{23}^2 + R_6 \sin(\theta_{46})\, \dot{\theta}_{46}^2 + R_8 \sin(\theta_8)\, \dot{\theta}_8^2$$

$$-R_7 \sin(\theta_7)\, \dot{\theta}_7^2 - R_5 \sin(\theta_{36} + \beta)\, \dot{\theta}_{36}^2 - R_3 \sin(\theta_{14} + \alpha)\, \dot{\theta}_{14}^2$$

For a range of 0º to 120º:

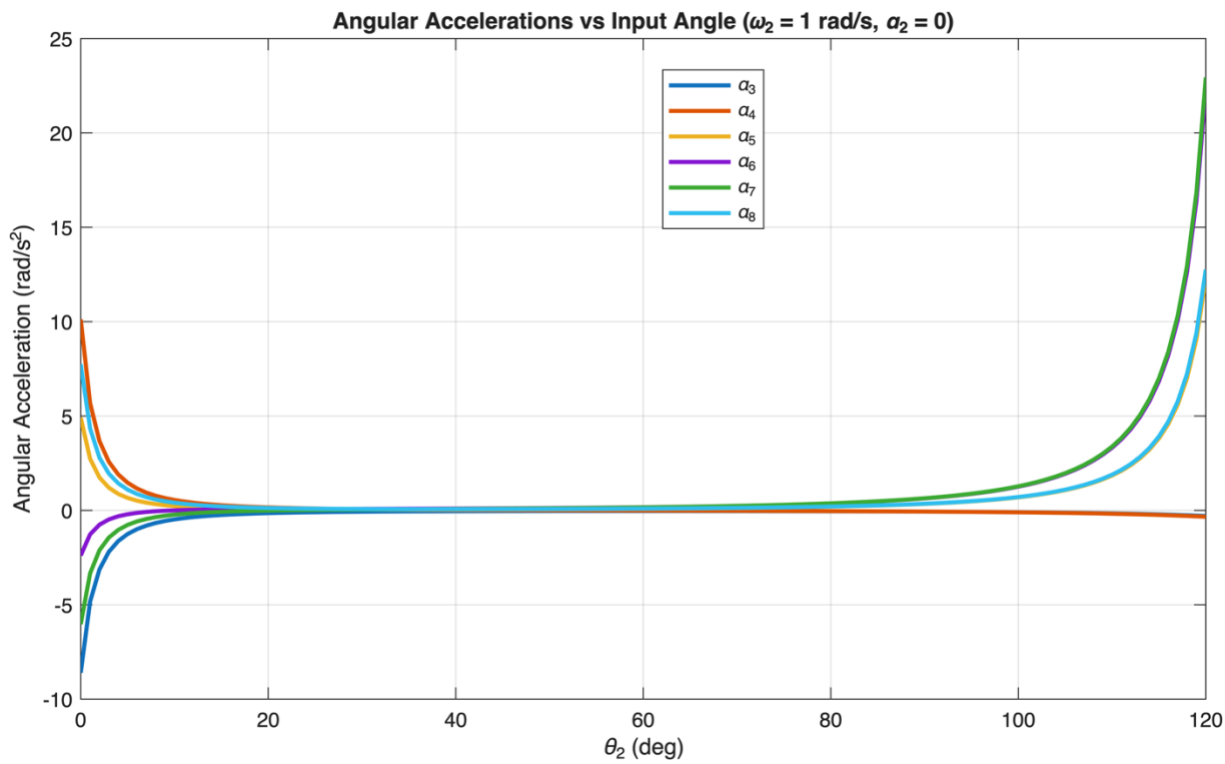


Figure 8: Angular Acceleration vs. Input Angle

The tabular results are presented in the table below. Complete table in appendix.

Table 8: Angular Acceleration at Select Input Angles (rad/s2)

θ_2 [deg]	α_3	α_4	α_5	α_6	α_7	α_8
0	-8.60	10.13	4.92	-2.39	-6.03	7.75
30	-0.07	0.06	0.05	0.08	0.04	0.07
60	-0.03	-0.01	0.09	0.17	0.16	0.09
90	-0.06	-0.06	0.35	0.64	0.64	0.35
120	-0.30	-0.34	12.29	22.29	22.94	12.76

From the velocity results, it can be observed that ω_3 and ω_6 remain the largest angular velocities throughout the input range, staying near 1 rad/s. This indicates that links 3 and 6 rotate at roughly the same rate as the input link. The angular velocity ω_4 remains small and slightly negative, consistent with the near-constant value of θ_4 observed in the posture analysis.

Near θ_2 = 120°, the velocities and accelerations of links 5, 6, 7 and 8 increase rapidly in magnitude. This behaviour suggests the mechanism is approaching a singular configuration at the upper limit of its operating range, which is consistent with the physical motion of a convertible car top reaching its fully open position.

4.3 Linear Acceleration Analysis

To determine the horizontal and vertical acceleration of Pin A, the position equations for Pin A from Section 3c were differentiated twice with respect to time. The position of Pin A is given by summing the vectors along the kinematic chain from the origin:

$$X_a = R_2 cos\theta_2 + R_4 cos\theta_{23} + R_6 cos\theta_{46} - (R_8 + R_a) cos\theta_8$$

$$Y_a = R_2 sin\theta_2 + R_4 sin\theta_{23} + R_6 sin\theta_{46} - (R_8 + R_a) sin\theta_8$$

The first time derivative of these equations gives the linear velocity components of Pin A. Each term produces a tangential velocity contribution from the corresponding angular velocity:

Differentiating a second time yields the linear acceleration of Pin A. Each term splits into a centripetal component (proportional to $R \cdot cos\theta \cdot \omega^2$) and a tangential component (proportional to $R \cdot sin\theta \cdot \alpha$). With $\alpha_2 = 0$ and $\omega_2 = 1$ rad/s, the horizontal and vertical accelerations of Pin A are:

$$a_{ax} = -R_2 cos\theta_2 \cdot {\omega_2}^2 - R_4 cos\theta_{23} \cdot {\omega_{23}}^2 - R_4 sin\theta_{23} \cdot \alpha_{23} - R_6 cos\theta_{46} \cdot {\omega_{46}}^2 - R_6 sin\theta_{46} \cdot \alpha_{46} + (R_8 + R_a) cos\theta_8 \cdot {\omega_8}^2 + (R_8 + R_a) sin\theta_8 \cdot \alpha_8$$

$$a_{ax} = -R_2 sin\theta_2 \cdot {\omega_2}^2 - R_4 sin\theta_{23} \cdot {\omega_{23}}^2 + R_4 cos\theta_{23} \cdot \alpha_{23} - R_6 sin\theta_{46} \cdot {\omega_{46}}^2 + R_6 cos\theta_{46} \cdot \alpha_{46} + (R_8 + R_a) sin\theta_8 \cdot {\omega_8}^2 - (R_8 + R_a) cos\theta_8 \cdot \alpha_8$$

where all angular positions, velocities, and accelerations are taken from the results of Questions 3a, 4a, and 4b respectively. The negative sign on the (R_8+R_a) terms in the position equation reflects the direction convention established in the loop 3 vector equation, where R_8 and R_7 were assigned opposite signs to R_6 to select the correct physical assembly configuration.

These equations were evaluated at each 1° increment from 0° to 120° using the MATLAB code. The resulting horizontal and vertical accelerations of Pin A are plotted below and tabulated in the appendix. The acceleration magnitudes remain moderate through most of the input range but increase sharply near θ_2 = 120°, consistent with the near-singular behaviour observed in the angular velocity and acceleration results.

Table 9: Linear Acceleration at Select Input Angles (mm/s2)

θ_2 [deg]	α_x	α_y
0	-1944.35	-10919.40
30	-593.94	-494.87
60	-334.17	-735.33
90	58.52	-1147.78
120	4897.59	-16405.23

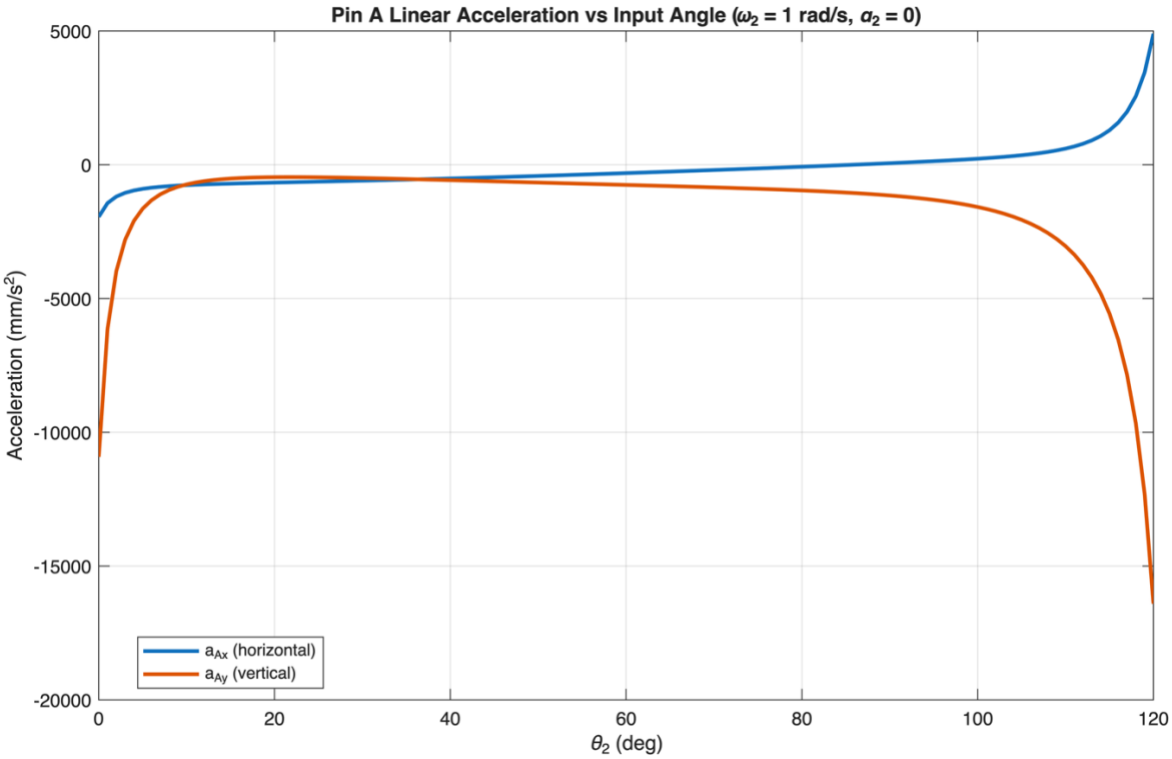


Figure 9: Linear Acceleration vs. Input Angle for Pin A

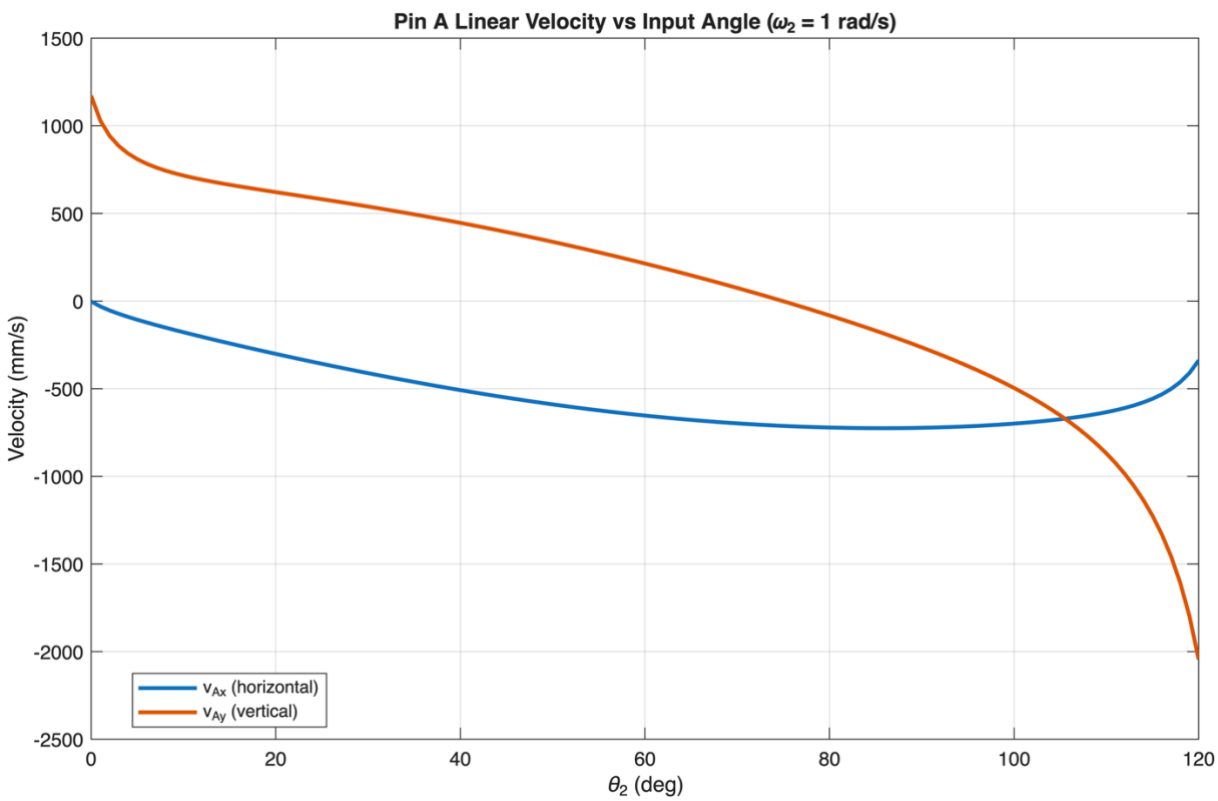
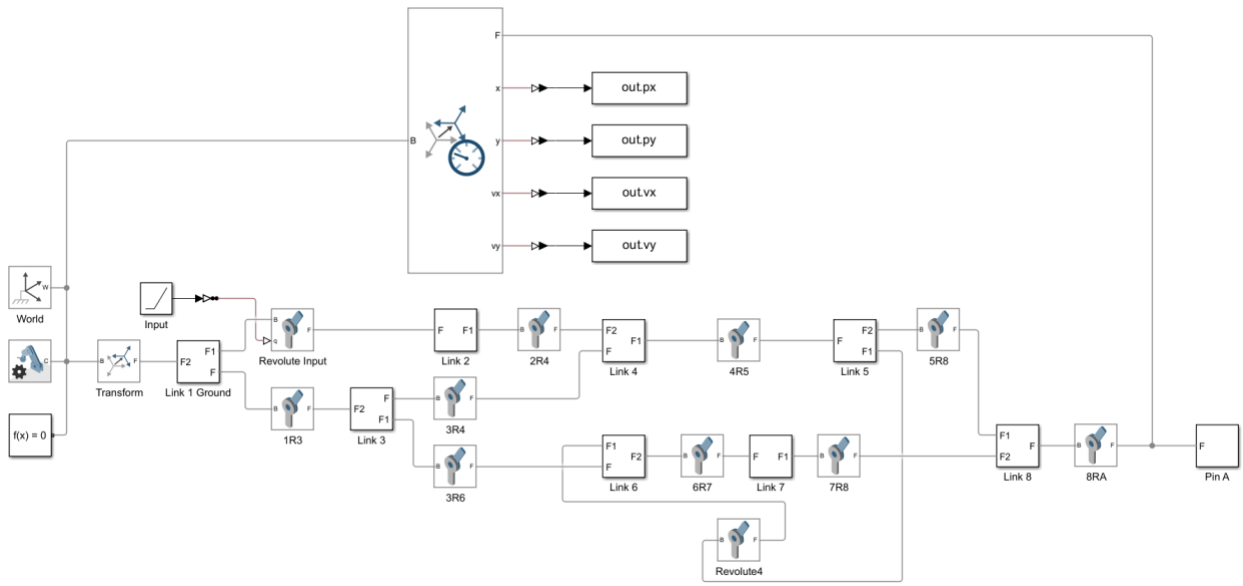


Figure 10: Linear Velocity vs. Input Angle for Pin A

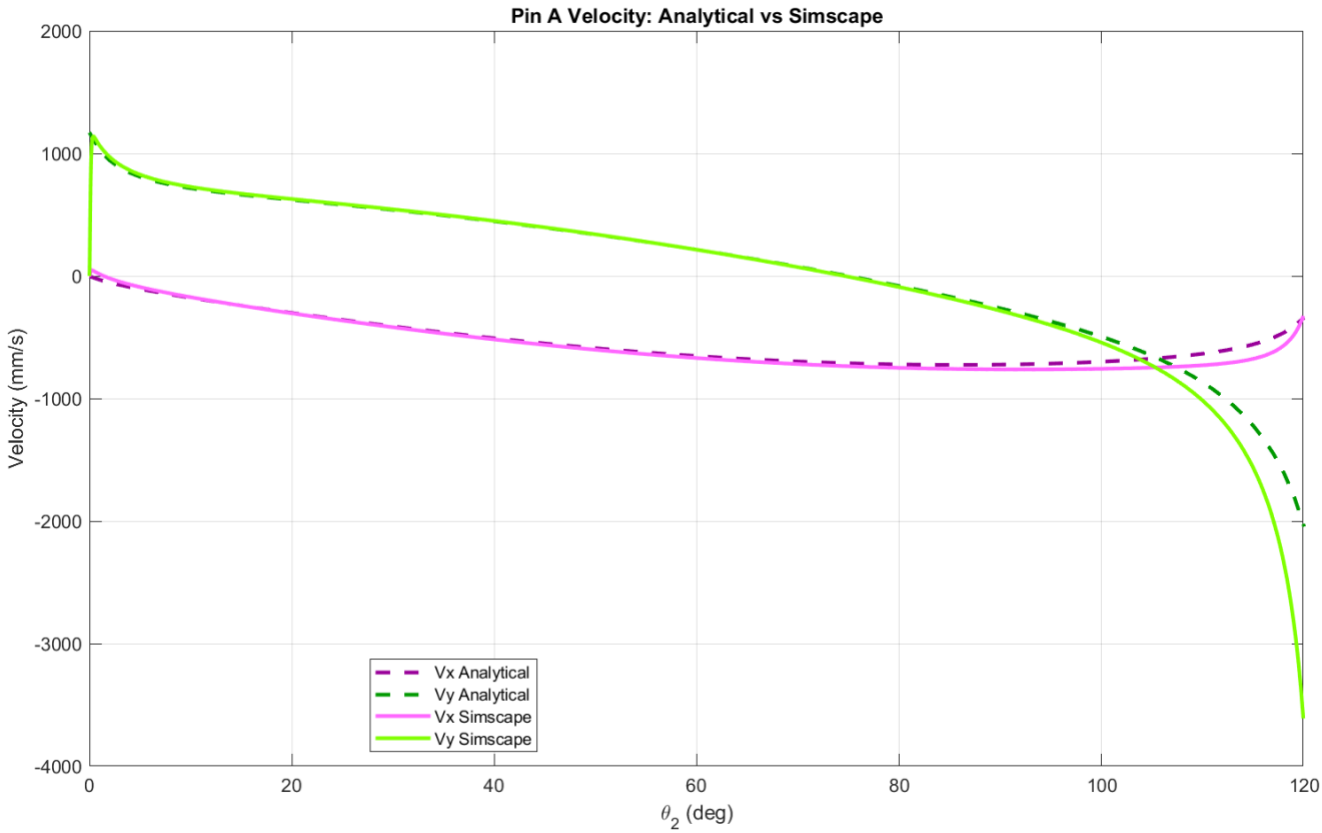
5 Simscape Analysis

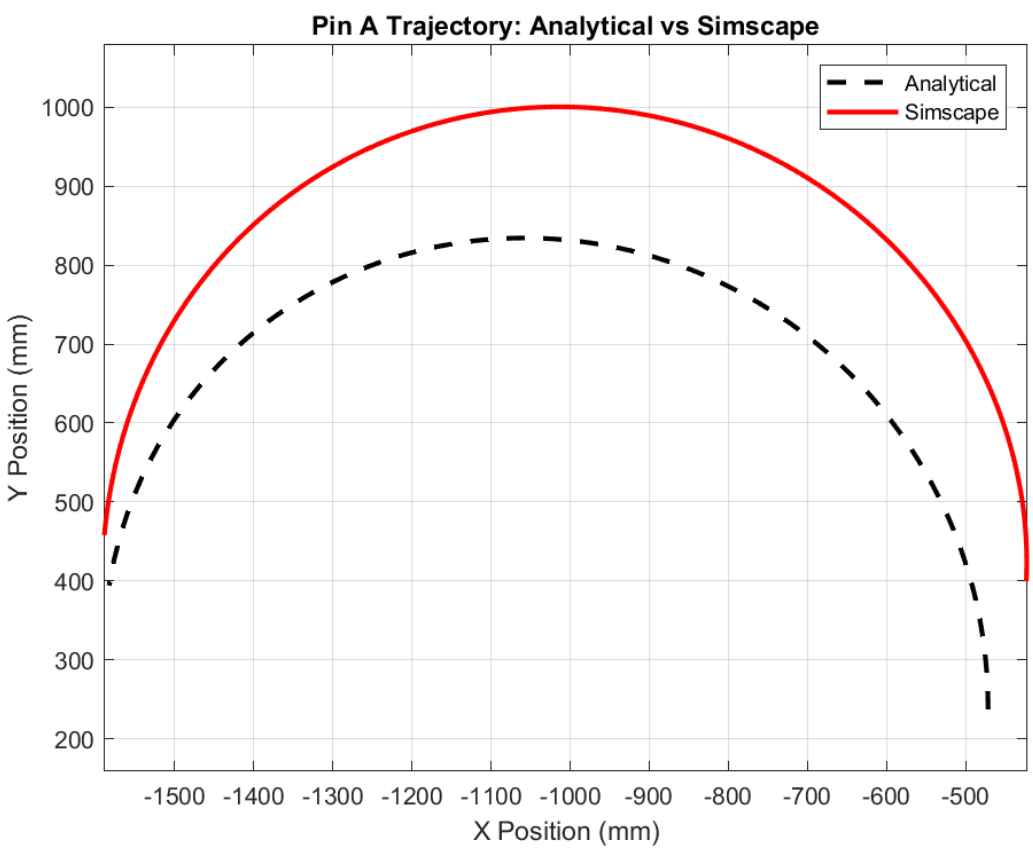
The convertible car top mechanism was modelled in SolidWorks and imported into MATLAB Simscape Multibody using an XML intermediate file. The Simscape model includes all eight links connected by revolute joints, with a prescribed constant angular velocity of 1 rad/s applied to the input link (link 2). Position and velocity outputs were extracted from the Simscape transform sensor block attached to the Pin A reference frame.

Simulation Workspace:



Results from simulation (Plots):


Pin A Velocity: Analytical vs Simscape
Velocity (mm/s)
θ_2 (deg)
Vx Analytical
Vy Analytical
Vx Simscape
Vy Simscape
2000
1000
0
-1000
-2000
-3000
-4000
0
20
40
60
80
100
120


Pin A Trajectory: Analytical vs Simscape
Analytical
Simscape
Y Position (mm)
X Position (mm)
1000
900
800
700
600
500
400
300
200
-1500
-1400
-1300
-1200
-1100
-1000
-900
-800
-700
-600
-500

5.1 Comparison of Pin A Trajectory

Figure 19 overlays the analytical and Simscape trajectories of Pin A for the full input range of 0° to 120°. The two curves follow the same arc shape, confirming that the Simscape rigid-body model is topologically consistent with the analytical vector-loop formulation. Both trajectories show Pin A sweeping from the lower-right region of the workspace near $\theta_2 = 0°$ to the upper-left near $\theta_2 = 60°–75°$, before descending to the lower-left at $\theta_2 = 120°$.

A vertical offset of approximately 100 mm is visible between the two trajectories. This discrepancy is attributed to differences in the coordinate reference frame origin between the SolidWorks/Simscape model and the analytical formulation. In the analytical model, the origin is placed at the right ground pivot (the intersection of links 1 and 2), whereas the Simscape model inherits its world frame from the SolidWorks assembly origin, which may be offset from this pin location. The consistent magnitude and direction of the offset across all input angles supports this explanation, as a simple frame translation would shift every point by the same vector without altering the curve shape.

5.2 Comparison of Pin A Velocity

Figure 20 compares the horizontal and vertical velocity components of Pin A from both the analytical and Simscape solutions. The analytical curves (dashed) and Simscape curves (solid) overlap closely throughout the operating range, confirming the correctness of the velocity analysis from Section 4. Minor deviations appear near $\theta_2 = 120°$, where the mechanism approaches a near-singular configuration and numerical sensitivity is highest. Overall, the close agreement between the two independent methods validates the Newton-Raphson kinematic solution and the Jacobian-based velocity computation.

6 Kinetics Analysis

6.1 Analytical Setup

To begin the kinetic analysis of the convertible rooftop system, a system of 21 equations was determined using the forces and moments acting on each link in the mechanism. The free body diagrams for each link can be seen below.

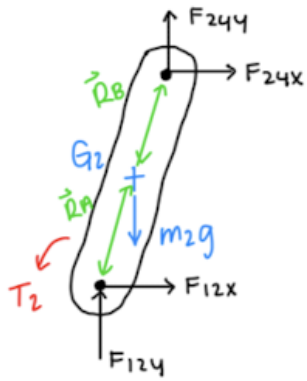


Figure 11: Free Body Diagram of Link 2

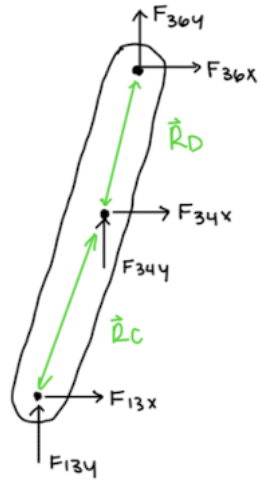


Figure 12: Free Body Diagram of Link 3

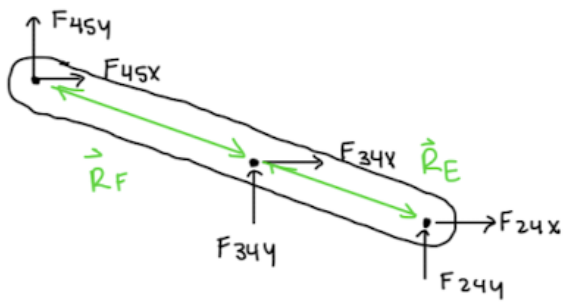


Figure 13: Free Body Diagram of Link 4

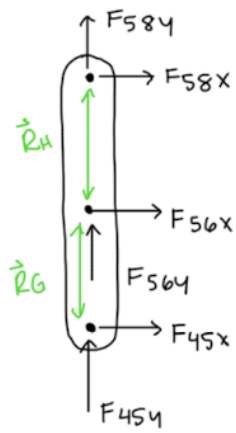


Figure 14: Free Body Diagram of Link 5

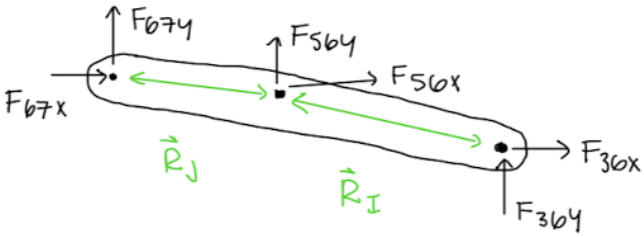


Figure 15: Free Body Diagram of Link 6

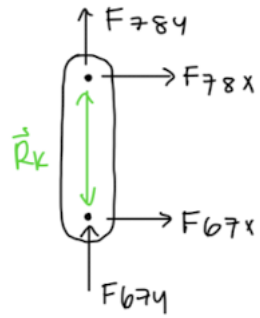


Figure 16: Free Body Diagram of Link 7

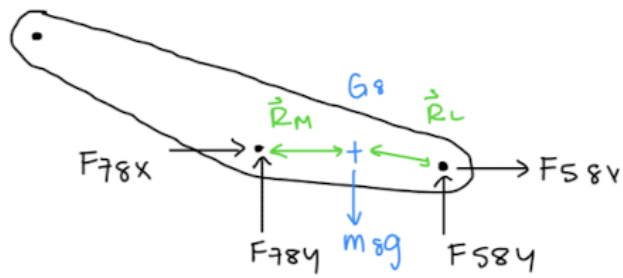


Figure 17: Free Body Diagram of Link 8

Three equations were then obtained for each link using force and moment summations. The equations can be seen below in Table 10.

Table 10: Kinetic Equations

Link 2	
$F_{12x} + F_{24x} = m_2 a_{G2x}$	(1)
$F_{12y} + F_{24y} = m_2 a_{G2y}$	(2)
$T_2 + R_{Ax}F_{12y} - R_{Ay}F_{12x} + R_{Bx}F_{24y} - R_{By}F_{24x} = I_{G2}\alpha_2$	(3)
Unknowns: $F_{12x}, F_{12y}, F_{24x}, F_{24y}, T_2$	
Link 3	
$F_{13x} + F_{34x} + F_{36x} = 0$	(4)
$F_{13y} + F_{34y} + F_{36y} = 0$	(5)
$R_{Cx}F_{13y} - R_{Cy}F_{13x} + R_{Dx}F_{36y} - R_{Dy}F_{36x} = 0$	(6)
Unknowns: $F_{13x}, F_{13y}, F_{34x}, F_{34y}, F_{36x}, F_{36y}$	
Link 4	
$F_{24x} + F_{34x} + F_{45x} = 0$	(7)
$F_{24y} + F_{34y} + F_{45y} = 0$	(8)
$R_{Ex}F_{24y} - R_{Ey}F_{24x} + R_{Fx}F_{45y} - R_{Fy}F_{45x} = 0$	(9)
Unknowns: F_{45x}, F_{45y}	
Link 5	
$F_{45x} + F_{56x} + F_{58x} = 0$	(10)
$F_{45y} + F_{56y} + F_{58y} = 0$	(11)
$R_{Gx}F_{45y} - R_{Gy}F_{45x} + R_{Hx}F_{58y} - R_{Hy}F_{58x} = 0$	(12)
Unknowns: $F_{56x}, F_{56y}, F_{58x}, F_{58y}$	
Link 6	
$F_{36x} + F_{56x} + F_{67x} = 0$	(13)
$F_{36y} + F_{56y} + F_{67y} = 0$	(14)
$R_{Ix}F_{36y} - R_{Iy}F_{36x} + R_{Jx}F_{67y} - R_{Jy}F_{67x} = 0$	(15)
Unknowns: F_{67x}, F_{67y}	
Link 7	

$F_{67x} + F_{78x} = 0$	(16)
$F_{67y} + F_{78y} = 0$	(17)
$R_{Kx}F_{78y} - R_{Ky}F_{78x} = 0$	(18)
Unknowns: F_{78x}, F_{78y}	
Link 8	
$F_{58x} + F_{78x} = m_8 a_{G8x}$	(19)
$F_{58y} + F_{78y} = m_8 a_{G8y}$	(20)
$R_{Lx}F_{58y} - R_{Ly}F_{58x} + R_{Mx}F_{78y} - R_{My}F_{78x} = I_{G8}\alpha_8$	(21)

This gives a total of 21 unknowns to go with the 21 equations. These equations can then be used in the following matrix to obtain the solution.

$$
\begin{bmatrix}
1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 1 & 0 & 0 & 0 & 1 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 1 & 0 & 0 & 0 & 1 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 1 & 0 & 1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 1 & 0 & 1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 1 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 1 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 1 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 1 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 1 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 1 & 0 \\
-R_{AY} & R_{AX} & 0 & 0 & -R_{BY} & R_{BX} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\
0 & 0 & -R_{CY} & R_{CX} & 0 & 0 & 0 & 0 & -R_{DY} & R_{DX} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & -R_{EY} & R_{EX} & 0 & 0 & 0 & 0 & -R_{FY} & R_{FX} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -R_{HY} & R_{HX} & 0 & 0 & -R_{GY} & R_{GX} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -R_{IY} & R_{IX} & 0 & 0 & 0 & 0 & 0 & 0 & -R_{JY} & R_{JX} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -R_{KY} & R_{KX} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -R_{LY} & R_{LX} & 0 & 0 & -R_{MY} & R_{MX} & 0
\end{bmatrix}
\begin{bmatrix}
F_{12x} \\ F_{12y} \\ F_{13x} \\ F_{13y} \\ F_{24x} \\ F_{24y} \\ F_{34x} \\ F_{34y} \\ F_{36x} \\ F_{36y} \\ F_{45x} \\ F_{45y} \\ F_{56x} \\ F_{56y} \\ F_{58x} \\ F_{58y} \\ F_{67x} \\ F_{67y} \\ F_{78x} \\ F_{78y} \\ T_2
\end{bmatrix}
=
\begin{bmatrix}
m_2 a_{G2x} \\ m_2 a_{G2y} \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ m_8 a_{G8x} \\ m_8 a_{G8y} \\ I_{G2}\alpha_2 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ I_{G8}\alpha_8
\end{bmatrix}
$$

6.2 Torque Requirements

The power equation (instantaneous power balance) states:

$$T_2\omega_2 = \dot{K}_{total} + P_{gravity}$$

where $\dot{K}$ is the rate of change of total kinetic energy and $P_{gravity}$ is the power done against gravity.

Conditions for the power equation to be valid:

1. All joint reactions are at frictionless pin joints thus they do no work
2. The only external loads are gravity (conservative) and the driving torque T_2.
3. There is one degree of freedom, so T_2 is the only unknown after kinematics are solved.

Under no friction, no aerodynamic forces, only gravity and T_2, the power equation gives T_2 directly from kinematics and mass properties, without needing to solve all internal joint forces.

The driving torque T_2 is plotted against the input angle θ_2 in Figure 11 and presented in tabular form at 10° increments in Table 12. The full tabulated results at every 1° increment are included in Appendix A.

Table 11: Variable Inputs for Mass and Length

Parameter	Link 2 (input crank)	Link 8 (output link)
Mass m (kg)	0.3474	1.1541
Length R (mm)	272.4	254.4
I_G (kg·m²)	2.14×10^{-4}	6.22×10^{-4}

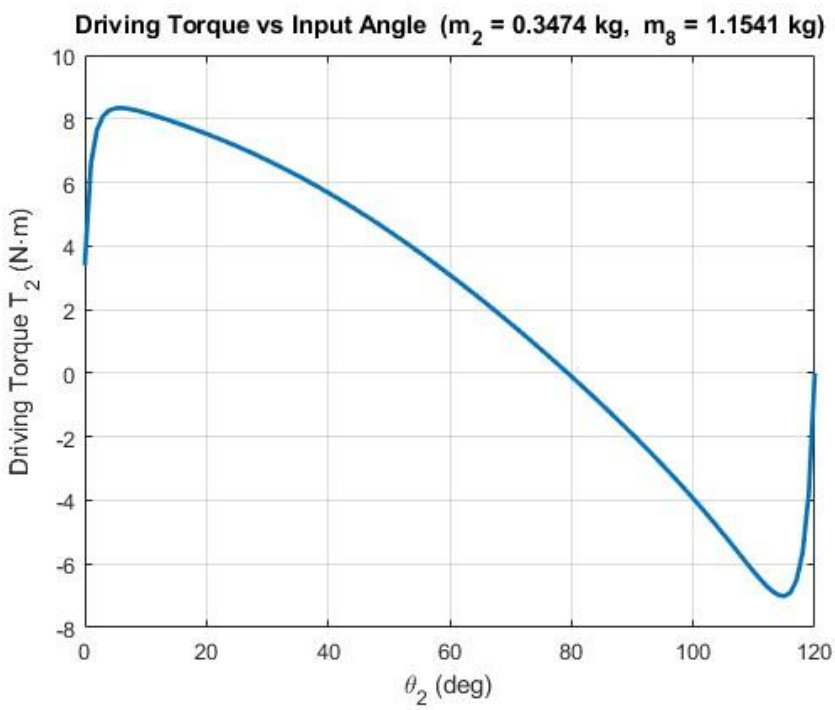


Figure 18: Driving Torque vs Input Angle

Table 12: Responding Torque Values for Major Theta Values

θ_2 [degrees]	T_2 [Nm]
0	3.39
30	6.69
60	3.09
90	-1.92
120	0.004

The driving torque T_2 remains positive throughout the operating range confirms that the actuator must continuously supply work to open the convertible roof against gravity. The torque is small near $\theta_2 = 0°$ because both links 2 and 8 are near-horizontal at the start of motion, so their centers of gravity have small vertical velocity components and gravity does little work per unit time.

As θ_2 increases through the mid-range, the roof is lifted through its highest elevation and the gravitational power term $m_8 g\dot{y}_{G8}$ is at its largest, producing the broad plateau visible in Figure 11. Near $\theta_2 = 120°$, both the angular velocity ω_8 and angular acceleration α_8 of link 8 increase sharply (approaching the singular configuration noted in Section 4), causing the inertia terms $\dot{K}_8$ to grow rapidly and T_2 to rise steeply at the end of the stroke.

A negative T_2 would indicate that gravity is driving the mechanism, and the actuator must resist the motion. For this mechanism and these mass values, T_2 remains positive throughout, meaning the actuator supplies positive work at every posture. This is physically consistent with a mechanism that lifts a roof against gravity over its entire operating range.

7 Peer Evaluation

7.1 Ratings

Table 13: Peer Evaluation Rubric Deliverable 1

	Mark	Cayden	Adele	Fadi
Mark		5	5	5
Cayden	5		5	5
Adele	5	5		5
Fadi	5	5	5	

Table 14: Peer Evaluation Rubric Deliverable 2

	Mark	Cayden	Adele	Fadi
Mark		5	5	5
Cayden	5		5	5
Adele	5	5		5
Fadi	5	5	5	

7.2 Contribution Statements

7.2.1 Mark

Deliverable 1: Took part in the initial setup of the deliverable (part 1), primarily focusing on labelling the links, joints and consequently discovering the mobility of the mechanism. Afterwards, worked on producing vector loop (part 2), assigning each three loops with the appropriate vector equations and constraints. Partook in debugging the MATLAB code for part 3 and formatting the word document.

Deliverable 2: Contributed to the optimization of the SolidWorks model; this includes ensuring defined sketches, proper mates, and that coordinate systems are present. Integrated the Solid Bodies into an XML file to be imported into Simscape Multibody. Plotted the posture and velocity of the simulation, followed by comparing it to the analytical values.

7.2.2 Cayden

Deliverable 1: Initially helped the team set up the governing loops and equations for question 1. Following this worked closely with Adele to complete question 3-part A and B whilst writing and polishing the report for submission. Work was completed ahead of schedule due to good communication by the group.

Deliverable 2: Complete question 6bi in order to prove that the power equation could be used to relate the torque needed to rotate the input link a certain amount. This was done through the extension of the problem statement 1's MATLAB code with results in section 6.2 of the report.

7.2.3 Adele

Deliverable 1: Completed the initial formatting for the report when the project was first released. Assisted in finalizing vector loops and equations to ensure links were consistently labelled throughout the different drawings. Completed the code for 3a and analytical solutions for 3b.

Deliverable 2: Completed the free body diagrams for each link in the mechanism, wrote the associated equations and verified that there were the same number of equations as there were unknowns and made the equation matrix for 6a. Completed section 6.1 in the report.

7.2.4 Fadi

Deliverable 1: At the start, I worked closely with the rest of the team on coming to a consensus for equation setup (question 1). Completed all of question 4, and part c for question 3, after communicating difficulties with team members regarding approach to solving it. Also updated some work for question 3, correcting inconsistencies in defining direction vectors for 7 and 8.

Deliverable 2: Created the CAD files in SolidWorks, ensuring that the linkages, joints, and mates were within constraints. Contributed to analysis of question 5, working with other team members to lay out the question, and did the write up for it. Did a complete read-over of the questions, ensuring all work is consistent.

8 Appendix A: Tabulated Results

8.1 Results for Question 3a

Table 15: Fully Tabulated Results for Question 3a from 0 to 120 Degrees

Theta 2 [º]	Theta 3 [º]	Theta 4 [º]	Theta 5 [º]	Theta 6 [º]	Theta 7 [º]	Theta 8 [º]
0	-0.75	167.48	168.90	10.73	8.72	175.19
1	0.61	167.02	168.76	11.85	10.02	174.92
2	1.88	166.66	168.67	12.95	11.25	174.72
3	3.09	166.37	168.61	14.04	12.46	174.58
4	4.27	166.12	168.57	15.12	13.63	174.47
5	5.42	165.91	168.55	16.19	14.79	174.38
6	6.55	165.72	168.54	17.26	15.93	174.31
7	7.66	165.56	168.54	18.33	17.07	174.26
8	8.75	165.41	168.54	19.39	18.20	174.22
9	9.84	165.27	168.55	20.46	19.32	174.19
10	10.91	165.15	168.57	21.52	20.43	174.17
11	11.98	165.03	168.59	22.59	21.54	174.16
12	13.03	164.92	168.61	23.65	22.65	174.15
13	14.09	164.82	168.64	24.72	23.75	174.15
14	15.13	164.73	168.67	25.79	24.86	174.15
15	16.17	164.64	168.70	26.85	25.96	174.16
16	17.21	164.56	168.73	27.92	27.06	174.17
17	18.25	164.48	168.77	28.99	28.16	174.18
18	19.28	164.41	168.81	30.06	29.26	174.20
19	20.30	164.33	168.84	31.13	30.36	174.22
20	21.33	164.26	168.88	32.20	31.45	174.24
21	22.35	164.20	168.92	33.28	32.55	174.26
22	23.37	164.14	168.97	34.35	33.65	174.29

23	24.39	164.07	169.01	35.43	34.75	174.32
24	25.40	164.01	169.05	36.50	35.85	174.35
25	26.42	163.96	169.10	37.58	36.94	174.38
26	27.43	163.90	169.14	38.66	38.04	174.41
27	28.44	163.85	169.19	39.74	39.14	174.44
28	29.45	163.79	169.24	40.82	40.24	174.48
29	30.46	163.74	169.29	41.91	41.34	174.52
30	31.46	163.69	169.33	42.99	42.44	174.55
31	32.47	163.64	169.38	44.08	43.55	174.59
32	33.47	163.60	169.44	45.17	44.65	174.63
33	34.48	163.55	169.49	46.25	45.75	174.68
34	35.48	163.50	169.54	47.34	46.86	174.72
35	36.48	163.46	169.59	48.44	47.96	174.76
36	37.48	163.41	169.65	49.53	49.07	174.81
37	38.48	163.37	169.70	50.62	50.18	174.86
38	39.48	163.32	169.76	51.72	51.29	174.90
39	40.48	163.28	169.81	52.82	52.40	174.95
40	41.47	163.24	169.87	53.92	53.51	175.00
41	42.47	163.20	169.93	55.02	54.62	175.05
42	43.46	163.15	169.98	56.12	55.73	175.11
43	44.46	163.11	170.04	57.22	56.85	175.16
44	45.45	163.07	170.10	58.33	57.97	175.21
45	46.45	163.03	170.17	59.44	59.09	175.27
46	47.44	162.99	170.23	60.55	60.21	175.33
47	48.43	162.95	170.29	61.66	61.33	175.38
48	49.42	162.91	170.36	62.77	62.45	175.44
49	50.41	162.87	170.42	63.89	63.58	175.50
50	51.40	162.83	170.49	65.01	64.70	175.56
51	52.39	162.79	170.55	66.13	65.83	175.63
52	53.38	162.75	170.62	67.25	66.96	175.69
53	54.37	162.71	170.69	68.37	68.10	175.76
54	55.36	162.67	170.76	69.50	69.23	175.82
55	56.35	162.63	170.83	70.63	70.37	175.89
56	57.33	162.59	170.91	71.76	71.51	175.96
57	58.32	162.56	170.98	72.89	72.65	176.03
58	59.30	162.52	171.06	74.03	73.80	176.10
59	60.29	162.48	171.13	75.17	74.95	176.18
60	61.27	162.44	171.21	76.31	76.10	176.25
61	62.26	162.40	171.29	77.46	77.25	176.33
62	63.24	162.36	171.37	78.60	78.40	176.41

63	64.22	162.32	171.46	79.75	79.56	176.49
64	65.21	162.28	171.54	80.91	80.73	176.57
65	66.19	162.23	171.63	82.07	81.89	176.65
66	67.17	162.19	171.72	83.23	83.06	176.74
67	68.15	162.15	171.81	84.39	84.23	176.83
68	69.13	162.11	171.90	85.56	85.41	176.92
69	70.11	162.07	171.99	86.73	86.59	177.01
70	71.09	162.03	172.09	87.91	87.77	177.10
71	72.06	161.98	172.19	89.09	88.96	177.20
72	73.04	161.94	172.29	90.27	90.15	177.30
73	74.02	161.90	172.39	91.46	91.35	177.40
74	74.99	161.85	172.50	92.65	92.55	177.50
75	75.97	161.81	172.61	93.85	93.75	177.61
76	76.94	161.76	172.72	95.06	94.97	177.72
77	77.92	161.72	172.83	96.27	96.18	177.83
78	78.89	161.67	172.95	97.48	97.41	177.95
79	79.86	161.62	173.07	98.70	98.63	178.07
80	80.84	161.58	173.19	99.93	99.87	178.19
81	81.81	161.53	173.32	101.16	101.11	178.32
82	82.78	161.48	173.45	102.40	102.36	178.44
83	83.75	161.43	173.58	103.65	103.61	178.58
84	84.71	161.38	173.72	104.91	104.87	178.72
85	85.68	161.33	173.86	106.17	106.15	178.86
86	86.65	161.28	174.01	107.44	107.42	179.01
87	87.61	161.22	174.16	108.72	108.71	179.16
88	88.58	161.17	174.32	110.01	110.01	179.31
89	89.54	161.12	174.48	111.31	111.32	179.48
90	90.50	161.06	174.65	112.62	112.64	179.64
91	91.46	161.00	174.82	113.94	113.96	179.82
92	92.43	160.95	175.00	115.27	115.31	180.00
93	93.38	160.89	175.19	116.62	116.66	180.19
94	94.34	160.83	175.39	117.98	118.03	180.38
95	95.30	160.76	175.59	119.35	119.41	180.59
96	96.25	160.70	175.80	120.74	120.81	180.80
97	97.21	160.64	176.02	122.14	122.22	181.02
98	98.16	160.57	176.25	123.56	123.65	181.25
99	99.11	160.50	176.49	125.00	125.10	181.49
100	100.06	160.43	176.74	126.46	126.57	181.74
101	101.01	160.36	177.00	127.95	128.06	182.01
102	101.95	160.29	177.28	129.45	129.58	182.29

103	102.90	160.21	177.57	130.99	131.13	182.58
104	103.84	160.14	177.87	132.55	132.70	182.89
105	104.78	160.06	178.20	134.14	134.30	183.22
106	105.72	159.97	178.54	135.77	135.94	183.56
107	106.65	159.89	178.90	137.43	137.62	183.93
108	107.58	159.80	179.29	139.13	139.34	184.32
109	108.52	159.71	179.70	140.88	141.10	184.74
110	109.44	159.62	180.14	142.69	142.92	185.18
111	110.37	159.53	180.61	144.55	144.79	185.66
112	111.29	159.43	181.12	146.47	146.74	186.18
113	112.21	159.33	181.67	148.47	148.76	186.74
114	113.13	159.22	182.27	150.56	150.87	187.35
115	114.04	159.11	182.92	152.75	153.08	188.02
116	114.95	159.00	183.65	155.06	155.42	188.75
117	115.85	158.88	184.45	157.52	157.90	189.57
118	116.75	158.75	185.35	160.15	160.56	190.49
119	117.65	158.63	186.38	163.00	163.45	191.54
120	118.54	158.49	187.56	166.14	166.64	192.75

8.2 Results for Question 3C – Pin A Position

Theta 2 [º]	X_A [mm]	Y_A [mm]
0.00	-471.50	237.11
1.00	-471.80	256.19
2.00	-472.54	273.34
3.00	-473.64	289.25
4.00	-475.07	304.30
5.00	-476.78	318.70
6.00	-478.78	332.60
7.00	-481.03	346.09
8.00	-483.54	359.25
9.00	-486.28	372.11
10.00	-489.27	384.73
11.00	-492.49	397.12
12.00	-495.93	409.30
13.00	-499.60	421.30
14.00	-503.49	433.13
15.00	-507.60	444.79
16.00	-511.92	456.30
17.00	-516.45	467.65

18.00	-521.20	478.86
19.00	-526.15	489.92
20.00	-531.30	500.84
21.00	-536.66	511.62
22.00	-542.22	522.26
23.00	-547.98	532.76
24.00	-553.94	543.11
25.00	-560.08	553.33
26.00	-566.42	563.41
27.00	-572.96	573.34
28.00	-579.67	583.12
29.00	-586.58	592.76
30.00	-593.66	602.25
31.00	-600.93	611.59
32.00	-608.38	620.78
33.00	-616.00	629.81
34.00	-623.80	638.68
35.00	-631.77	647.40
36.00	-639.90	655.95
37.00	-648.21	664.34
38.00	-656.67	672.56
39.00	-665.30	680.61
40.00	-674.09	688.48
41.00	-683.03	696.18
42.00	-692.13	703.71
43.00	-701.38	711.05
44.00	-710.77	718.21
45.00	-720.31	725.18
46.00	-730.00	731.96
47.00	-739.82	738.56
48.00	-749.78	744.95
49.00	-759.87	751.16
50.00	-770.09	757.16
51.00	-780.44	762.96
52.00	-790.92	768.55
53.00	-801.51	773.94
54.00	-812.22	779.11
55.00	-823.05	784.07
56.00	-833.99	788.82
57.00	-845.04	793.35

58.00	-856.19	797.65
59.00	-867.45	801.74
60.00	-878.80	805.59
61.00	-890.25	809.22
62.00	-901.79	812.61
63.00	-913.42	815.77
64.00	-925.13	818.70
65.00	-936.93	821.38
66.00	-948.80	823.82
67.00	-960.75	826.01
68.00	-972.76	827.95
69.00	-984.85	829.64
70.00	-997.00	831.08
71.00	-1009.20	832.26
72.00	-1021.47	833.18
73.00	-1033.79	833.84
74.00	-1046.15	834.22
75.00	-1058.57	834.34
76.00	-1071.02	834.19
77.00	-1083.51	833.76
78.00	-1096.03	833.04
79.00	-1108.59	832.05
80.00	-1121.17	830.76
81.00	-1133.78	829.18
82.00	-1146.40	827.31
83.00	-1159.04	825.13
84.00	-1171.69	822.65
85.00	-1184.34	819.86
86.00	-1197.00	816.75
87.00	-1209.66	813.32
88.00	-1222.31	809.56
89.00	-1234.95	805.47
90.00	-1247.58	801.03
91.00	-1260.20	796.24
92.00	-1272.79	791.10
93.00	-1285.35	785.59
94.00	-1297.88	779.70
95.00	-1310.38	773.43
96.00	-1322.83	766.75
97.00	-1335.24	759.66

98.00	-1347.60	752.15
99.00	-1359.90	744.19
100.00	-1372.14	735.77
101.00	-1384.31	726.87
102.00	-1396.41	717.47
103.00	-1408.42	707.54
104.00	-1420.34	697.05
105.00	-1432.17	685.97
106.00	-1443.88	674.26
107.00	-1455.48	661.89
108.00	-1466.94	648.79
109.00	-1478.26	634.92
110.00	-1489.41	620.20
111.00	-1500.38	604.55
112.00	-1511.14	587.88
113.00	-1521.66	570.06
114.00	-1531.90	550.96
115.00	-1541.82	530.38
116.00	-1551.33	508.11
117.00	-1560.37	483.83
118.00	-1568.80	457.16
119.00	-1576.44	427.51
120.00	-1583.01	394.08

8.3 Results for Question 4b- Velocity

Table 16: Fully Tabulated Data for Question 4B from 0 to 120 Degrees

Theta 2	omega 3	omega 4	omega 5	omega 6	omega 7	omega 8
0	1.4205	-0.5318	-0.1744	1.1407	1.3424	-0.3292
1	1.3082	-0.3996	-0.1104	1.1102	1.2642	-0.2282
2	1.2404	-0.3200	-0.0724	1.0931	1.2179	-0.1675
3	1.1948	-0.2663	-0.0471	1.0825	1.1875	-0.1268
4	1.1618	-0.2277	-0.0292	1.0757	1.1662	-0.0976
5	1.1370	-0.1985	-0.0159	1.0713	1.1506	-0.0757
6	1.1175	-0.1757	-0.0057	1.0684	1.1389	-0.0586
7	1.1019	-0.1575	0.0023	1.0666	1.1299	-0.0450
8	1.0891	-0.1426	0.0088	1.0656	1.1229	-0.0339
9	1.0785	-0.1302	0.0141	1.0650	1.1173	-0.0247
10	1.0695	-0.1197	0.0184	1.0649	1.1129	-0.0170
11	1.0618	-0.1108	0.0221	1.0651	1.1094	-0.0104

12	1.0551	-0.1032	0.0253	1.0655	1.1065	-0.0047
13	1.0493	-0.0965	0.0280	1.0661	1.1043	0.0002
14	1.0442	-0.0907	0.0304	1.0668	1.1024	0.0046
15	1.0397	-0.0856	0.0325	1.0676	1.1010	0.0084
16	1.0357	-0.0811	0.0343	1.0685	1.0999	0.0118
17	1.0321	-0.0771	0.0360	1.0695	1.0991	0.0149
18	1.0289	-0.0735	0.0375	1.0705	1.0985	0.0177
19	1.0260	-0.0703	0.0388	1.0716	1.0980	0.0202
20	1.0233	-0.0674	0.0401	1.0727	1.0978	0.0226
21	1.0209	-0.0648	0.0412	1.0738	1.0977	0.0247
22	1.0187	-0.0624	0.0423	1.0750	1.0978	0.0267
23	1.0166	-0.0602	0.0434	1.0762	1.0979	0.0285
24	1.0148	-0.0583	0.0443	1.0774	1.0982	0.0303
25	1.0130	-0.0565	0.0452	1.0787	1.0985	0.0319
26	1.0114	-0.0549	0.0461	1.0799	1.0990	0.0334
27	1.0099	-0.0534	0.0470	1.0812	1.0995	0.0349
28	1.0085	-0.0520	0.0478	1.0825	1.1001	0.0363
29	1.0072	-0.0508	0.0486	1.0839	1.1008	0.0376
30	1.0059	-0.0496	0.0494	1.0852	1.1015	0.0389
31	1.0047	-0.0486	0.0502	1.0866	1.1023	0.0402
32	1.0036	-0.0476	0.0510	1.0880	1.1032	0.0414
33	1.0026	-0.0467	0.0518	1.0894	1.1041	0.0426
34	1.0016	-0.0459	0.0526	1.0909	1.1051	0.0438
35	1.0006	-0.0452	0.0533	1.0923	1.1061	0.0449
36	0.9997	-0.0445	0.0541	1.0938	1.1072	0.0461
37	0.9989	-0.0439	0.0549	1.0954	1.1083	0.0472
38	0.9980	-0.0433	0.0557	1.0969	1.1095	0.0483
39	0.9973	-0.0428	0.0565	1.0985	1.1108	0.0494
40	0.9965	-0.0423	0.0574	1.1001	1.1121	0.0505
41	0.9957	-0.0419	0.0582	1.1018	1.1134	0.0516
42	0.9950	-0.0415	0.0591	1.1035	1.1148	0.0527
43	0.9943	-0.0411	0.0599	1.1052	1.1163	0.0538
44	0.9937	-0.0408	0.0608	1.1070	1.1178	0.0550
45	0.9930	-0.0406	0.0617	1.1088	1.1194	0.0561
46	0.9923	-0.0403	0.0627	1.1107	1.1211	0.0572
47	0.9917	-0.0401	0.0636	1.1126	1.1228	0.0584
48	0.9911	-0.0399	0.0646	1.1146	1.1245	0.0596
49	0.9905	-0.0398	0.0656	1.1166	1.1264	0.0608
50	0.9899	-0.0397	0.0667	1.1187	1.1283	0.0620
51	1.4205	-0.5318	-0.1744	1.1407	1.1302	0.0633

52	1.3082	-0.3996	-0.1104	1.1102	1.1323	0.0646
53	1.2404	-0.3200	-0.0724	1.0931	1.1344	0.0659
54	1.1948	-0.2663	-0.0471	1.0825	1.1366	0.0672
55	1.1618	-0.2277	-0.0292	1.0757	1.1389	0.0686
56	1.1370	-0.1985	-0.0159	1.0713	1.1412	0.0700
57	1.1175	-0.1757	-0.0057	1.0684	1.1437	0.0714
58	1.1019	-0.1575	0.0023	1.0666	1.1462	0.0729
59	1.0891	-0.1426	0.0088	1.0656	1.1488	0.0745
60	1.0785	-0.1302	0.0141	1.0650	1.1516	0.0761
61	1.0695	-0.1197	0.0184	1.0649	1.1544	0.0777
62	1.0618	-0.1108	0.0221	1.0651	1.1574	0.0794
63	1.0551	-0.1032	0.0253	1.0655	1.1604	0.0811
64	1.0493	-0.0965	0.0280	1.0661	1.1636	0.0830
65	1.0442	-0.0907	0.0304	1.0668	1.1669	0.0848
66	1.0397	-0.0856	0.0325	1.0676	1.1704	0.0868
67	1.0357	-0.0811	0.0343	1.0685	1.1740	0.0888
68	1.0321	-0.0771	0.0360	1.0695	1.1777	0.0909
69	1.0289	-0.0735	0.0375	1.0705	1.1816	0.0930
70	1.0260	-0.0703	0.0388	1.0716	1.1857	0.0953
71	1.0233	-0.0674	0.0401	1.0727	1.1899	0.0977
72	1.0209	-0.0648	0.0412	1.0738	1.1943	0.1001
73	1.0187	-0.0624	0.0423	1.0750	1.1989	0.1027
74	1.0166	-0.0602	0.0434	1.0762	1.2037	0.1053
75	1.0148	-0.0583	0.0443	1.0774	1.2088	0.1081
76	1.0130	-0.0565	0.0452	1.0787	1.2140	0.1110
77	1.0114	-0.0549	0.0461	1.0799	1.2195	0.1141
78	1.0099	-0.0534	0.0470	1.0812	1.2253	0.1172
79	1.0085	-0.0520	0.0478	1.0825	1.2314	0.1206
80	1.0072	-0.0508	0.0486	1.0839	1.2377	0.1241
81	1.0059	-0.0496	0.0494	1.0852	1.2444	0.1278
82	1.0047	-0.0486	0.0502	1.0866	1.2514	0.1316
83	1.0036	-0.0476	0.0510	1.0880	1.2588	0.1357
84	1.0026	-0.0467	0.0518	1.0894	1.2666	0.1400
85	1.0016	-0.0459	0.0526	1.0909	1.2748	0.1445
86	1.0006	-0.0452	0.0533	1.0923	1.2835	0.1492
87	0.9997	-0.0445	0.0541	1.0938	1.2926	0.1543
88	0.9989	-0.0439	0.0549	1.0954	1.3023	0.1596
89	0.9980	-0.0433	0.0557	1.0969	1.3125	0.1652
90	0.9973	-0.0428	0.0565	1.0985	1.3234	0.1712
91	0.9965	-0.0423	0.0574	1.1001	1.3350	0.1776

92	0.9957	-0.0419	0.0582	1.1018	1.3472	0.1844
93	0.9950	-0.0415	0.0591	1.1035	1.3603	0.1916
94	0.9943	-0.0411	0.0599	1.1052	1.3743	0.1993
95	0.9937	-0.0408	0.0608	1.1070	1.3892	0.2075
96	0.9930	-0.0406	0.0617	1.1088	1.4051	0.2163
97	0.9923	-0.0403	0.0627	1.1107	1.4223	0.2258
98	0.9917	-0.0401	0.0636	1.1126	1.4407	0.2360
99	0.9911	-0.0399	0.0646	1.1146	1.4605	0.2470
100	0.9905	-0.0398	0.0656	1.1166	1.4819	0.2589
101	0.9899	-0.0397	0.0667	1.1187	1.5051	0.2718
102	1.4205	-0.5318	-0.1744	1.1407	1.5303	0.2857
103	1.3082	-0.3996	-0.1104	1.1102	1.5577	0.3010
104	1.2404	-0.3200	-0.0724	1.0931	1.5876	0.3176
105	1.1948	-0.2663	-0.0471	1.0825	1.6205	0.3359
106	1.1618	-0.2277	-0.0292	1.0757	1.6566	0.3561
107	1.1370	-0.1985	-0.0159	1.0713	1.6965	0.3784
108	1.1175	-0.1757	-0.0057	1.0684	1.7409	0.4031
109	1.1019	-0.1575	0.0023	1.0666	1.7904	0.4308
110	1.0891	-0.1426	0.0088	1.0656	1.8459	0.4619
111	1.0785	-0.1302	0.0141	1.0650	1.9088	0.4970
112	1.0695	-0.1197	0.0184	1.0649	1.9803	0.5371
113	1.0618	-0.1108	0.0221	1.0651	2.0626	0.5832
114	1.0551	-0.1032	0.0253	1.0655	2.1581	0.6367
115	1.0493	-0.0965	0.0280	1.0661	2.2702	0.6995
116	1.0442	-0.0907	0.0304	1.0668	2.4040	0.7744
117	1.0397	-0.0856	0.0325	1.0676	2.5664	0.8653
118	1.0357	-0.0811	0.0343	1.0685	2.7679	0.9780
119	1.0321	-0.0771	0.0360	1.0695	3.0256	1.1218
120	1.0289	-0.0735	0.0375	1.0705	3.3685	1.3128

8.4 Results for Question 4b – Acceleration

Table 17: Fully Tabulated Data for Question 4B

Theta 2	Alpha 3	Alpha 4	Alpha 5	Alpha 6	Alpha 7	Alpha 8
0	-8.6049	10.1258	4.9190	-2.3890	-6.0320	7.7457
1	-4.8327	5.6839	2.7336	-1.2680	-3.3343	4.3374
2	-3.1232	3.6708	1.7416	-0.7558	-2.1073	2.7909
3	-2.1918	2.5738	1.2021	-0.4778	-1.4388	1.9485
4	-1.6243	1.9053	0.8750	-0.3107	-1.0328	1.4360

5	-1.2514	1.4660	0.6616	-0.2033	-0.7674	1.1001
6	-0.9928	1.1611	0.5150	-0.1309	-0.5845	0.8678
7	-0.8058	0.9406	0.4102	-0.0804	-0.4535	0.7006
8	-0.6662	0.7759	0.3331	-0.0441	-0.3565	0.5764
9	-0.5592	0.6496	0.2749	-0.0176	-0.2830	0.4817
10	-0.4754	0.5506	0.2301	0.0021	-0.2261	0.4080
11	-0.4087	0.4717	0.1950	0.0170	-0.1812	0.3497
12	-0.3547	0.4077	0.1672	0.0284	-0.1452	0.3029
13	-0.3104	0.3552	0.1450	0.0373	-0.1160	0.2648
14	-0.2737	0.3116	0.1269	0.0442	-0.0920	0.2335
15	-0.2429	0.2749	0.1122	0.0496	-0.0721	0.2076
16	-0.2169	0.2439	0.1001	0.0540	-0.0554	0.1859
17	-0.1948	0.2174	0.0901	0.0575	-0.0412	0.1676
18	-0.1758	0.1946	0.0818	0.0604	-0.0290	0.1521
19	-0.1594	0.1748	0.0749	0.0628	-0.0185	0.1389
20	-0.1452	0.1576	0.0691	0.0648	-0.0094	0.1276
21	-0.1327	0.1425	0.0643	0.0666	-0.0013	0.1179
22	-0.1218	0.1292	0.0602	0.0681	0.0058	0.1096
23	-0.1122	0.1174	0.0569	0.0695	0.0122	0.1024
24	-0.1038	0.1069	0.0541	0.0708	0.0179	0.0962
25	-0.0962	0.0976	0.0518	0.0721	0.0232	0.0908
26	-0.0895	0.0892	0.0499	0.0733	0.0280	0.0861
27	-0.0835	0.0816	0.0483	0.0745	0.0324	0.0821
28	-0.0782	0.0747	0.0471	0.0756	0.0365	0.0786
29	-0.0734	0.0685	0.0462	0.0769	0.0404	0.0756
30	-0.0691	0.0629	0.0455	0.0781	0.0441	0.0731
31	-0.0652	0.0577	0.0450	0.0794	0.0476	0.0709
32	-0.0617	0.0529	0.0447	0.0807	0.0510	0.0690
33	-0.0585	0.0486	0.0446	0.0821	0.0543	0.0675
34	-0.0556	0.0446	0.0446	0.0836	0.0575	0.0662
35	-0.0530	0.0409	0.0448	0.0851	0.0607	0.0652
36	-0.0507	0.0374	0.0452	0.0867	0.0638	0.0644
37	-0.0485	0.0342	0.0456	0.0885	0.0669	0.0639
38	-0.0466	0.0312	0.0462	0.0903	0.0699	0.0635
39	-0.0448	0.0284	0.0469	0.0922	0.0730	0.0633
40	-0.0432	0.0258	0.0477	0.0942	0.0761	0.0633
41	-0.0418	0.0233	0.0486	0.0963	0.0793	0.0634
42	-0.0405	0.0210	0.0495	0.0985	0.0825	0.0637
43	-0.0393	0.0188	0.0506	0.1008	0.0857	0.0641
44	-0.0382	0.0167	0.0518	0.1033	0.0890	0.0647

45	-0.0373	0.0147	0.0531	0.1059	0.0924	0.0654
46	-0.0364	0.0128	0.0545	0.1086	0.0959	0.0663
47	-0.0357	0.0110	0.0559	0.1114	0.0994	0.0673
48	-0.0350	0.0092	0.0575	0.1144	0.1031	0.0684
49	-0.0344	0.0076	0.0592	0.1176	0.1069	0.0696
50	-0.0339	0.0059	0.0609	0.1209	0.1108	0.0710
51	-0.0335	0.0044	0.0628	0.1244	0.1149	0.0725
52	-0.0331	0.0029	0.0648	0.1281	0.1191	0.0742
53	-0.0328	0.0014	0.0669	0.1319	0.1235	0.0759
54	-0.0326	-0.0001	0.0691	0.1360	0.1280	0.0778
55	-0.0324	-0.0015	0.0714	0.1402	0.1327	0.0799
56	-0.0323	-0.0029	0.0738	0.1447	0.1377	0.0821
57	-0.0322	-0.0043	0.0764	0.1494	0.1428	0.0844
58	-0.0322	-0.0056	0.0791	0.1544	0.1482	0.0869
59	-0.0323	-0.0070	0.0820	0.1596	0.1538	0.0895
60	-0.0324	-0.0083	0.0850	0.1650	0.1597	0.0924
61	-0.0325	-0.0096	0.0882	0.1708	0.1658	0.0954
62	-0.0327	-0.0109	0.0915	0.1769	0.1722	0.0985
63	-0.0330	-0.0123	0.0950	0.1833	0.1790	0.1019
64	-0.0333	-0.0136	0.0988	0.1900	0.1861	0.1055
65	-0.0337	-0.0149	0.1027	0.1971	0.1936	0.1093
66	-0.0341	-0.0163	0.1068	0.2047	0.2014	0.1133
67	-0.0345	-0.0177	0.1112	0.2126	0.2097	0.1176
68	-0.0350	-0.0190	0.1158	0.2210	0.2184	0.1222
69	-0.0356	-0.0204	0.1207	0.2298	0.2276	0.1270
70	-0.0362	-0.0219	0.1259	0.2392	0.2373	0.1321
71	-0.0369	-0.0233	0.1314	0.2491	0.2476	0.1375
72	-0.0376	-0.0248	0.1372	0.2597	0.2584	0.1433
73	-0.0384	-0.0264	0.1433	0.2708	0.2699	0.1494
74	-0.0392	-0.0279	0.1499	0.2827	0.2821	0.1560
75	-0.0401	-0.0295	0.1568	0.2953	0.2950	0.1629
76	-0.0411	-0.0312	0.1642	0.3087	0.3088	0.1704
77	-0.0421	-0.0329	0.1721	0.3230	0.3234	0.1783
78	-0.0432	-0.0347	0.1805	0.3382	0.3390	0.1867
79	-0.0444	-0.0365	0.1895	0.3544	0.3556	0.1958
80	-0.0456	-0.0384	0.1991	0.3717	0.3733	0.2054
81	-0.0470	-0.0404	0.2094	0.3903	0.3922	0.2158
82	-0.0484	-0.0424	0.2204	0.4102	0.4125	0.2269
83	-0.0499	-0.0445	0.2322	0.4315	0.4343	0.2389
84	-0.0515	-0.0467	0.2449	0.4544	0.4576	0.2517

85	-0.0532	-0.0490	0.2585	0.4790	0.4827	0.2655
86	-0.0550	-0.0515	0.2733	0.5056	0.5098	0.2804
87	-0.0569	-0.0540	0.2892	0.5342	0.5389	0.2966
88	-0.0589	-0.0566	0.3064	0.5651	0.5705	0.3140
89	-0.0610	-0.0594	0.3250	0.5987	0.6046	0.3329
90	-0.0633	-0.0623	0.3452	0.6350	0.6416	0.3535
91	-0.0657	-0.0654	0.3673	0.6746	0.6819	0.3759
92	-0.0683	-0.0686	0.3913	0.7177	0.7257	0.4003
93	-0.0710	-0.0720	0.4175	0.7647	0.7736	0.4269
94	-0.0739	-0.0756	0.4462	0.8162	0.8260	0.4562
95	-0.0770	-0.0794	0.4778	0.8728	0.8835	0.4883
96	-0.0803	-0.0834	0.5125	0.9350	0.9468	0.5236
97	-0.0838	-0.0876	0.5509	1.0037	1.0167	0.5627
98	-0.0875	-0.0921	0.5934	1.0797	1.0941	0.6061
99	-0.0915	-0.0969	0.6406	1.1642	1.1801	0.6542
100	-0.0958	-0.1019	0.6933	1.2583	1.2759	0.7080
101	-0.1003	-0.1073	0.7523	1.3637	1.3832	0.7682
102	-0.1052	-0.1130	0.8186	1.4822	1.5038	0.8359
103	-0.1103	-0.1191	0.8935	1.6159	1.6400	0.9124
104	-0.1159	-0.1257	0.9785	1.7677	1.7946	0.9993
105	-0.1219	-0.1327	1.0755	1.9408	1.9710	1.0985
106	-0.1283	-0.1401	1.1868	2.1396	2.1736	1.2124
107	-0.1351	-0.1482	1.3155	2.3692	2.4077	1.3441
108	-0.1425	-0.1568	1.4651	2.6364	2.6803	1.4975
109	-0.1505	-0.1660	1.6407	2.9500	3.0004	1.6776
110	-0.1591	-0.1760	1.8486	3.3215	3.3798	1.8909
111	-0.1684	-0.1867	2.0972	3.7663	3.8343	2.1463
112	-0.1785	-0.1984	2.3981	4.3051	4.3853	2.4558
113	-0.1894	-0.2109	2.7674	4.9672	5.0630	2.8361
114	-0.2012	-0.2246	3.2280	5.7943	5.9104	3.3111
115	-0.2141	-0.2394	3.8135	6.8479	6.9911	3.9159
116	-0.2281	-0.2555	4.5753	8.2223	8.4026	4.7045
117	-0.2433	-0.2730	5.5956	10.0682	10.3015	5.7629
118	-0.2600	-0.2922	7.0127	12.6412	12.9534	7.2373
119	-0.2783	-0.3132	9.0776	16.4059	16.8427	9.3932
120	-0.2984	-0.3362	12.2899	22.2913	22.9413	12.7619

8.5 Results for Question 4C – Acceleration

Theta 2 [º]	a_X [mm/s²]	a_Y [mm/s²]
0.00	-1944.35	-10919.40
1.00	-1432.45	-6137.28
2.00	-1187.83	-3975.60
3.00	-1048.46	-2804.73
4.00	-960.10	-2097.90
5.00	-899.85	-1639.40
6.00	-856.48	-1326.54
7.00	-823.90	-1105.02
8.00	-798.52	-943.79
9.00	-778.14	-823.96
10.00	-761.31	-733.54
11.00	-747.05	-664.54
12.00	-734.70	-611.53
13.00	-723.76	-570.67
14.00	-713.89	-539.19
15.00	-704.82	-515.06
16.00	-696.35	-496.77
17.00	-688.34	-483.16
18.00	-680.66	-473.33
19.00	-673.22	-466.60
20.00	-665.96	-462.42
21.00	-658.80	-460.34
22.00	-651.71	-460.01
23.00	-644.64	-461.16
24.00	-637.57	-463.53
25.00	-630.47	-466.95
26.00	-623.33	-471.25
27.00	-616.11	-476.30
28.00	-608.82	-481.98
29.00	-601.43	-488.20
30.00	-593.94	-494.87
31.00	-586.35	-501.94
32.00	-578.64	-509.35
33.00	-570.81	-517.03
34.00	-562.85	-524.95
35.00	-554.78	-533.08
36.00	-546.57	-541.37

37.00	-538.23	-549.80
38.00	-529.77	-558.35
39.00	-521.17	-566.99
40.00	-512.44	-575.71
41.00	-503.58	-584.49
42.00	-494.59	-593.31
43.00	-485.47	-602.17
44.00	-476.22	-611.05
45.00	-466.84	-619.95
46.00	-457.34	-628.86
47.00	-447.71	-637.77
48.00	-437.96	-646.68
49.00	-428.09	-655.58
50.00	-418.10	-664.47
51.00	-407.99	-673.35
52.00	-397.77	-682.23
53.00	-387.43	-691.09
54.00	-376.99	-699.94
55.00	-366.44	-708.79
56.00	-355.79	-717.63
57.00	-345.03	-726.48
58.00	-334.17	-735.33
59.00	-323.22	-744.19
60.00	-312.17	-753.07
61.00	-301.02	-761.98
62.00	-289.79	-770.93
63.00	-278.47	-779.93
64.00	-267.07	-788.98
65.00	-255.58	-798.11
66.00	-244.01	-807.34
67.00	-232.36	-816.67
68.00	-220.63	-826.12
69.00	-208.83	-835.72
70.00	-196.95	-845.49
71.00	-185.01	-855.46
72.00	-172.99	-865.65
73.00	-160.89	-876.10
74.00	-148.73	-886.83
75.00	-136.50	-897.89
76.00	-124.19	-909.33

77.00	-111.81	-921.18
78.00	-99.35	-933.50
79.00	-86.82	-946.35
80.00	-74.20	-959.79
81.00	-61.49	-973.90
82.00	-48.69	-988.75
83.00	-35.79	-1004.43
84.00	-22.78	-1021.05
85.00	-9.64	-1038.71
86.00	3.63	-1057.54
87.00	17.06	-1077.70
88.00	30.67	-1099.33
89.00	44.48	-1122.62
90.00	58.52	-1147.78
91.00	72.83	-1175.04
92.00	87.45	-1204.69
93.00	102.44	-1237.03
94.00	117.85	-1272.42
95.00	133.77	-1311.28
96.00	150.28	-1354.10
97.00	167.49	-1401.45
98.00	185.55	-1453.99
99.00	204.62	-1512.51
100.00	224.91	-1577.94
101.00	246.67	-1651.40
102.00	270.21	-1734.25
103.00	295.93	-1828.09
104.00	324.33	-1934.93
105.00	356.05	-2057.22
106.00	391.90	-2198.00
107.00	432.95	-2361.12
108.00	480.59	-2551.45
109.00	536.67	-2775.29
110.00	603.69	-3040.88
111.00	685.08	-3359.18
112.00	785.64	-3745.05
113.00	912.22	-4219.15
114.00	1074.86	-4810.84
115.00	1288.70	-5563.23
116.00	1577.48	-6541.97

117.00	1979.99	-7851.53
118.00	2563.25	-9667.77
119.00	3451.49	-12308.49
120.00	4897.59	-16405.23

8.6 Results for Question 6b – i

Theta2 deg	Driving Torque Nm
0	3.3901
1	6.5688
2	7.6509
3	8.0841
4	8.2639
5	8.3298
6	8.3394
7	8.3192
8	8.2823
9	8.2357
10	8.183
11	8.1263
12	8.0668
13	8.005
14	7.9415
15	7.8763
16	7.8095
17	7.7411
18	7.6712
19	7.5996
20	7.5264
21	7.4516
22	7.375
23	7.2967
24	7.2165
25	7.1345
26	7.0507
27	6.9649
28	6.8773
29	6.7878
30	6.6963
31	6.6029
32	6.5075

33	6.4102
34	6.311
35	6.2098
36	6.1067
37	6.0017
38	5.8947
39	5.7858
40	5.6751
41	5.5624
42	5.4479
43	5.3315
44	5.2133
45	5.0933
46	4.9715
47	4.8478
48	4.7225
49	4.5953
50	4.4665
51	4.336
52	4.2037
53	4.0699
54	3.9343
55	3.7972
56	3.6585
57	3.5181
58	3.3763
59	3.2329
60	3.088
61	2.9416
62	2.7937
63	2.6443
64	2.4935
65	2.3413
66	2.1877
67	2.0327
68	1.8763
69	1.7186
70	1.5594
71	1.399
72	1.2371
73	1.074

74	0.9094
75	0.7436
76	0.5764
77	0.4078
78	0.2378
79	0.0665
80	-0.1062
81	-0.2804
82	-0.456
83	-0.6331
84	-0.8118
85	-0.992
86	-1.1739
87	-1.3574
88	-1.5428
89	-1.7299
90	-1.919
91	-2.1101
92	-2.3033
93	-2.4988
94	-2.6967
95	-2.897
96	-3.1
97	-3.3058
98	-3.5145
99	-3.7264
100	-3.9415
101	-4.16
102	-4.382
103	-4.6075
104	-4.8366
105	-5.0689
106	-5.3043
107	-5.5418
108	-5.7803
109	-6.0179
110	-6.2512
111	-6.4752
112	-6.6817
113	-6.8574
114	-6.9806

115	-7.0148
116	-6.8964
117	-6.5101
118	-5.6344
119	-3.812
120	0.0037